

Durham College
Information Systems Security – Computers and Networking
INFT 1103 Operating System Security

Lab 3: Installation of Nessus to scan

Lab Workbook

Name – Student ID	Course Code	Weight
Madhvi Lashkaree (100997946)	INFT 1103	5%

Part A: Install Windows server 2019 and Install Nessus on a Windows server 2019 Datacenter

Part B: Scan your Domain controller

Part C: Scan Report

Lab Objectives

Upon completion of this lab, you will be able to perform the following:

- Install Windows 2019 Virtual Machine (VM).
- Install Nessus vulnerability scanner
- Run a host discovery and vulnerability scan

Lab Materials

- **Please name your Lab workbook titled “Lab 2 - <name> <Student ID>”.**
 - Please include all relevant screenshots on Lab workbook for submission (provided on dc connect)
- Install Windows Server 2019 VM
- You will need your windows server 2016 domain controller installed from previous assignment

Lab Requirements

- VMware workstation 16 pro
- Windows server 2019
- Windows server 2016 (from Lab 1 and 2 assignment)
- Nessus installer

Nessus is a vulnerability scanner used for vulnerability management and penetrations Testing. Nessus scans a specified IP and checks for vulnerabilities that can be exploited by a threat actor. It does this scan by running several scripts called plugin id's on a target to check for known vulnerabilities.

Part A:

Install windows 2019 server and Set IP Address

1. Go to your Azure cloud to download windows server 2019
2. Download windows server 2019

[Instructor's screenshot]

Software
Education



Windows Server 2019 Datacenter (updated Sept 2019)

Windows Server 2019 Standard is ideal for customers with low density or non-virtualized environments. Windows Server 2019 offers additional features in Standard and Datacenter editions. Features exclusive to the Windows Server 2016 Datacenter edition include Shielded Virtual Machines, software-defined networking, Storage Spaces Direct, and Storage Replica. While no features from the Windows Server 2016 R2 Standard edition have been removed, we have added features like Nano Server and unlimited Windows Server containers to the Windows Server 2019 Standard edition.

Operating System

Windows

Product language

English

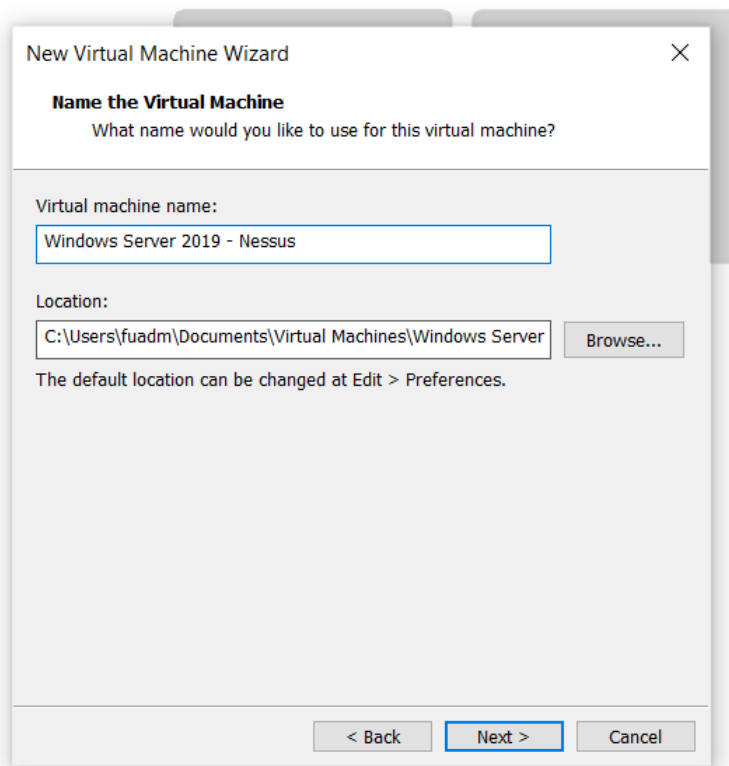
System

64 bit

[View Key](#)

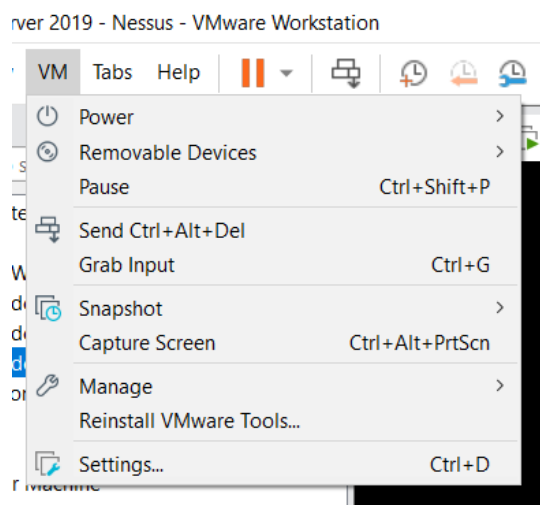
3. Install the windows 2019 server on your VMware (I believe you all should know how to install this by now, same way you installed the widows 2016 server)

[Instructor's screenshot]



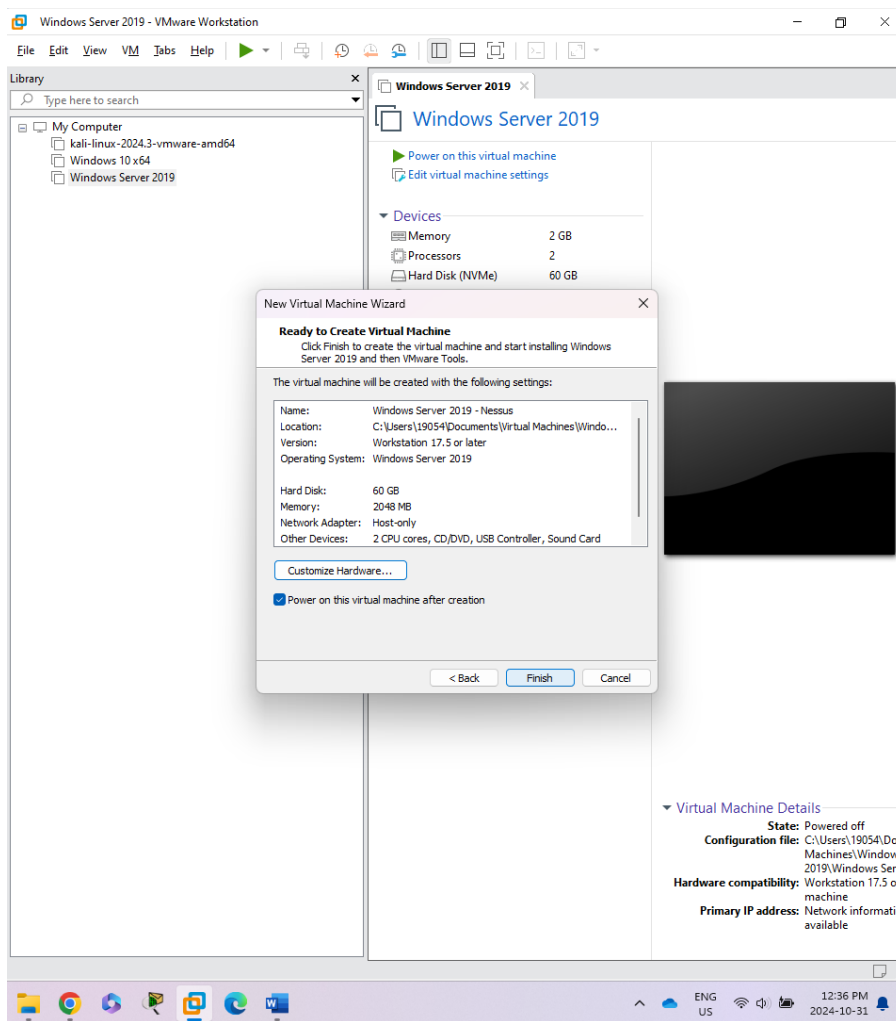
4. Go over to your installed windows 2019 server VM settings, go over to your VM settings by clicking VM > settings

[Instructor's screenshot]



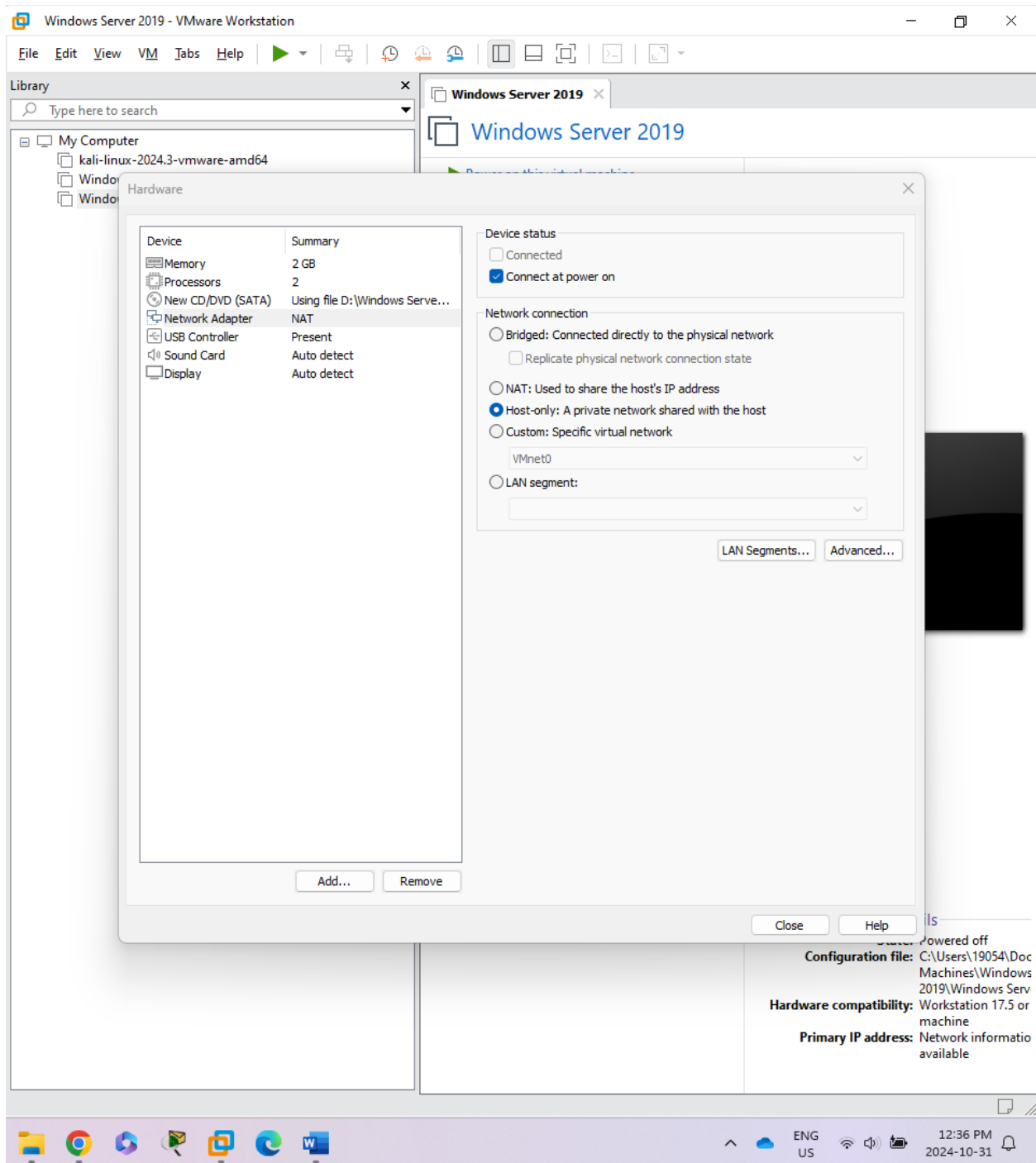
5. Create an additional network Adapter by clicking Add > Network Adapter > Finish > OK

[Attach your screenshot]



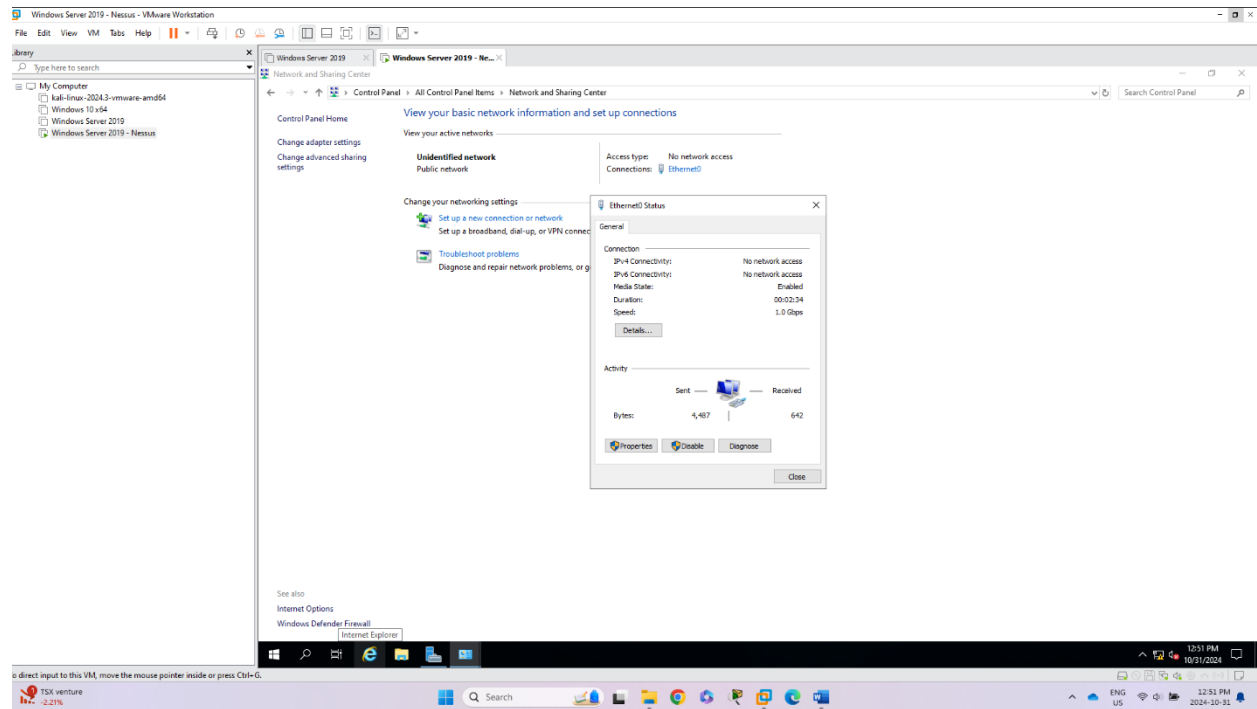
6. Set your Network adapter to NAT and Network Adapter 2 to Host only

[Attach your screenshot]



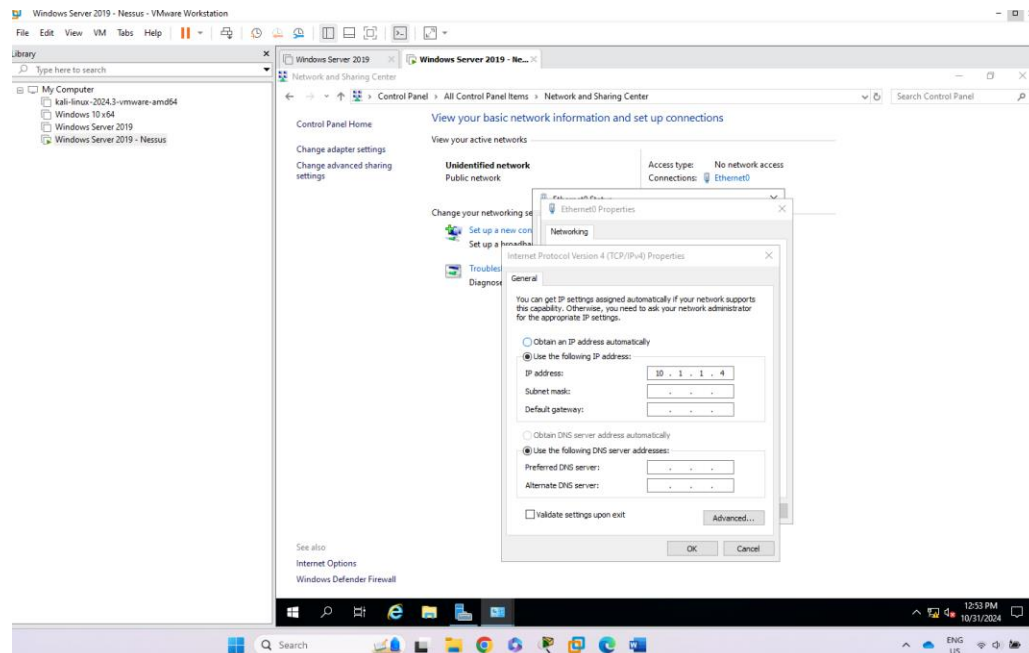
- Next step would be to go to control panel > Network sharing > Change adapter settings of your Windows 2019 server to change our IP address of Ethernet 1, double click on Ethernet 1

[Attach your screenshot]



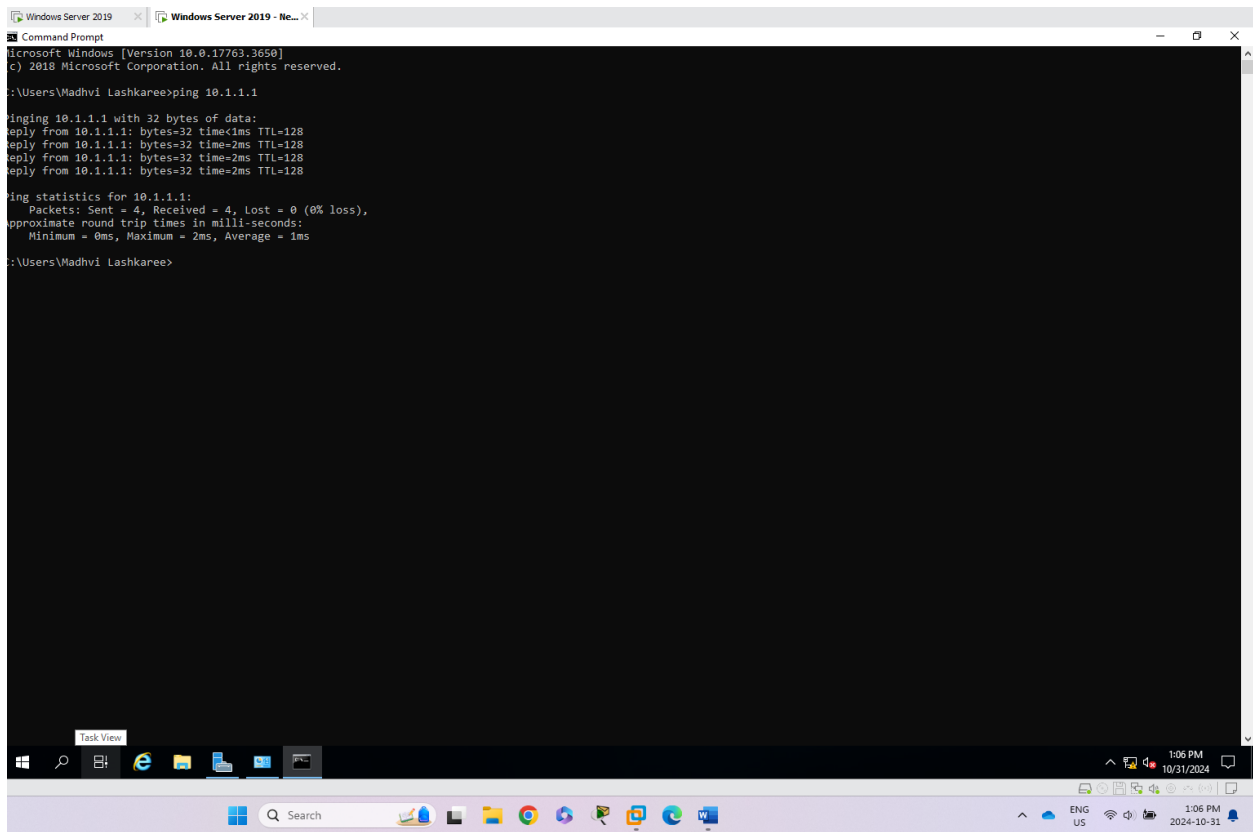
- Click on properties > IPv4 Address and set IP address to 10.1.1.4 as seen below

[Attach your screenshot]



9. Go over to the command prompt on your windows 2019 server and try to ping your windows 2016 Domain controller (if you are having problems with pinging, troubleshoot your network adapter settings or disable firewall on your windows 2016 server, if it does not work, troubleshoot and find how to fix it. **“PLEASE DO NOT EMAIL ME ASKING HOW TO PING YOUR MACHINES”**)

[Attach your screenshot]



```
Windows Server 2019 - Command Prompt
Microsoft Windows [Version 10.0.17763.3650]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Madhvi Lashkaree>ping 10.1.1.1

Pinging 10.1.1.1 with 32 bytes of data:
Reply from 10.1.1.1: bytes=32 time<1ms TTL=128
Reply from 10.1.1.1: bytes=32 time=2ms TTL=128
Reply from 10.1.1.1: bytes=32 time=2ms TTL=128
Reply from 10.1.1.1: bytes=32 time=2ms TTL=128

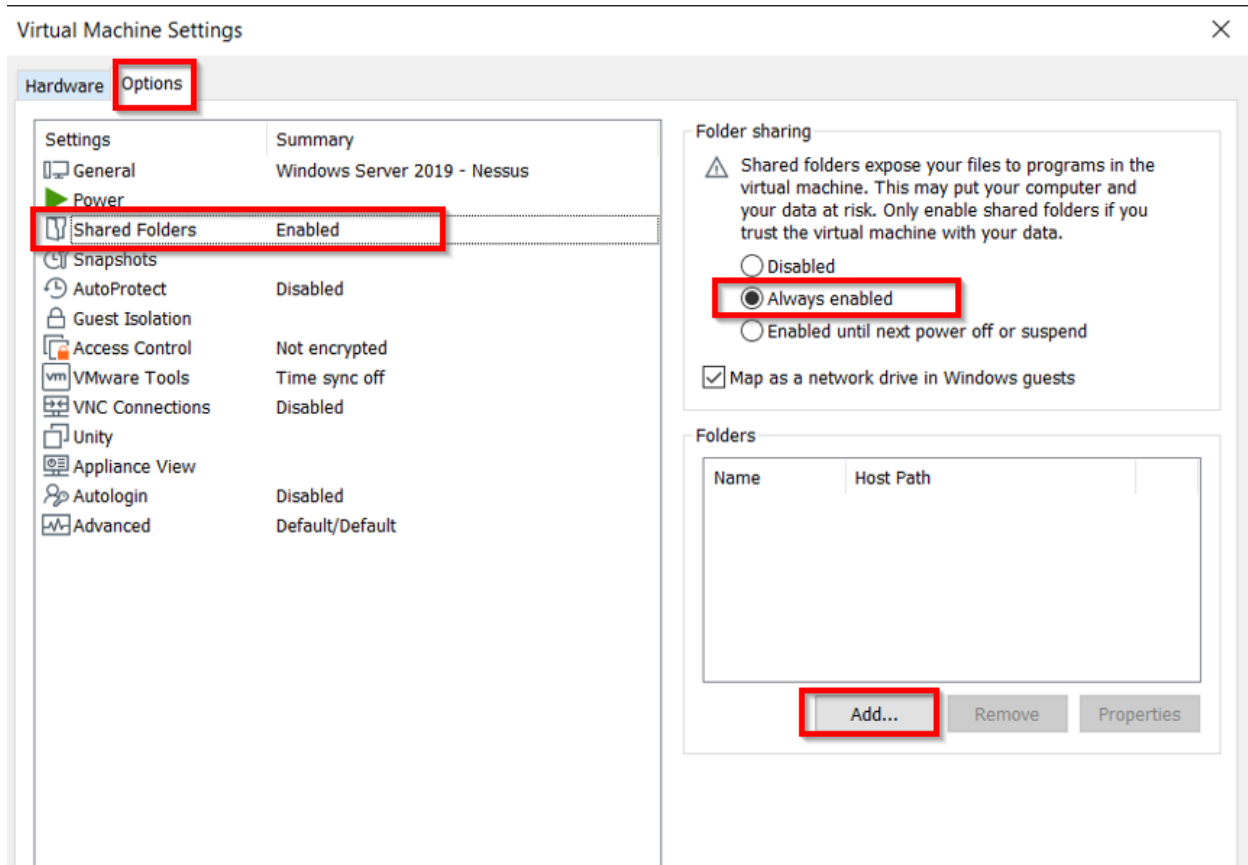
Ping statistics for 10.1.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\Users\Madhvi Lashkaree>
```

10. Your Windows 2019 server would be connected to the internet because we have set one of our network adapters to NAT.

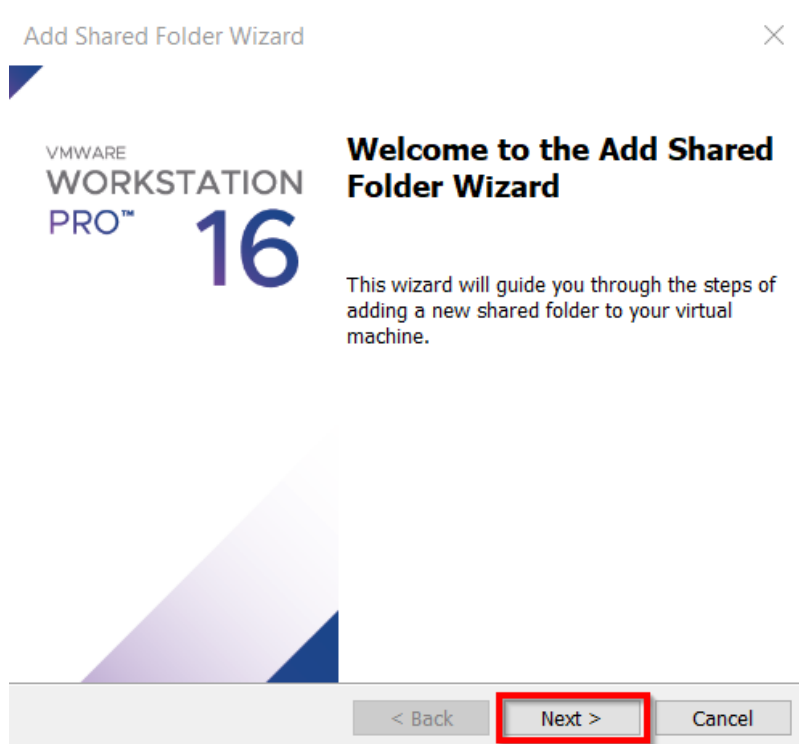
In your windows 2019 server, let us download and install chrome browser to make it comfortable for browsing. It would be difficult downloading chrome on internet explorer, so we would download it on our host machine and move it to the windows server 2019. To do this, we would create a share between the host machine and the windows server 2019. Go to VM > Settings and click on Options, shared folder, Always enabled and click Add

[Instructor's screenshot]



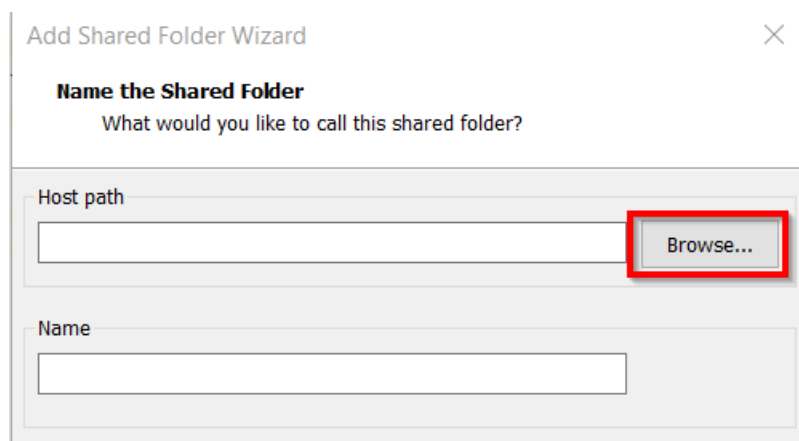
11. After clicking Add, a wizard would open. Then you click next

[Instructor's screenshot]



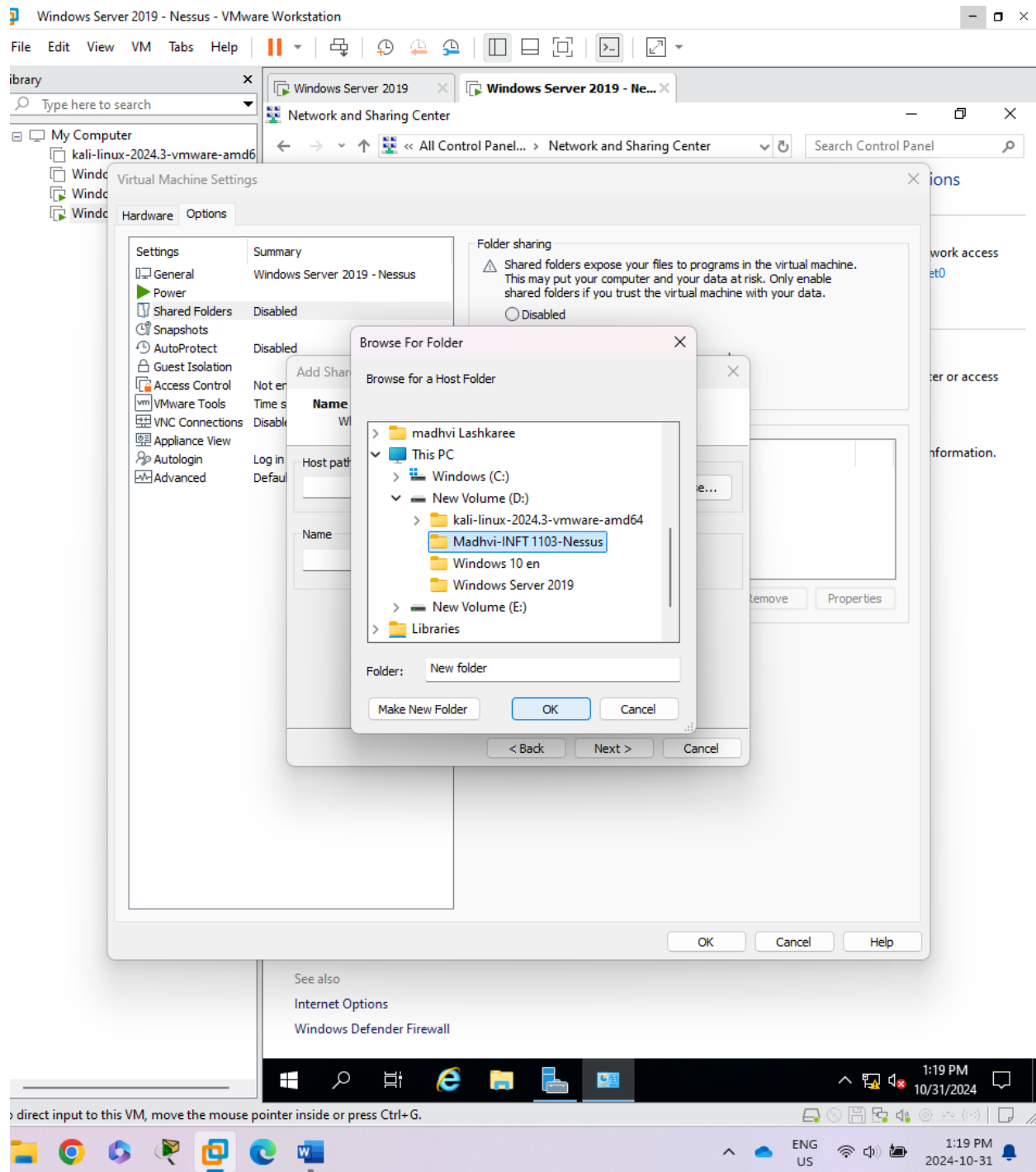
12. Next step would be to add the folder you would like to share and the path in your host machine. To do this, click browse

[Instructor's screenshot]



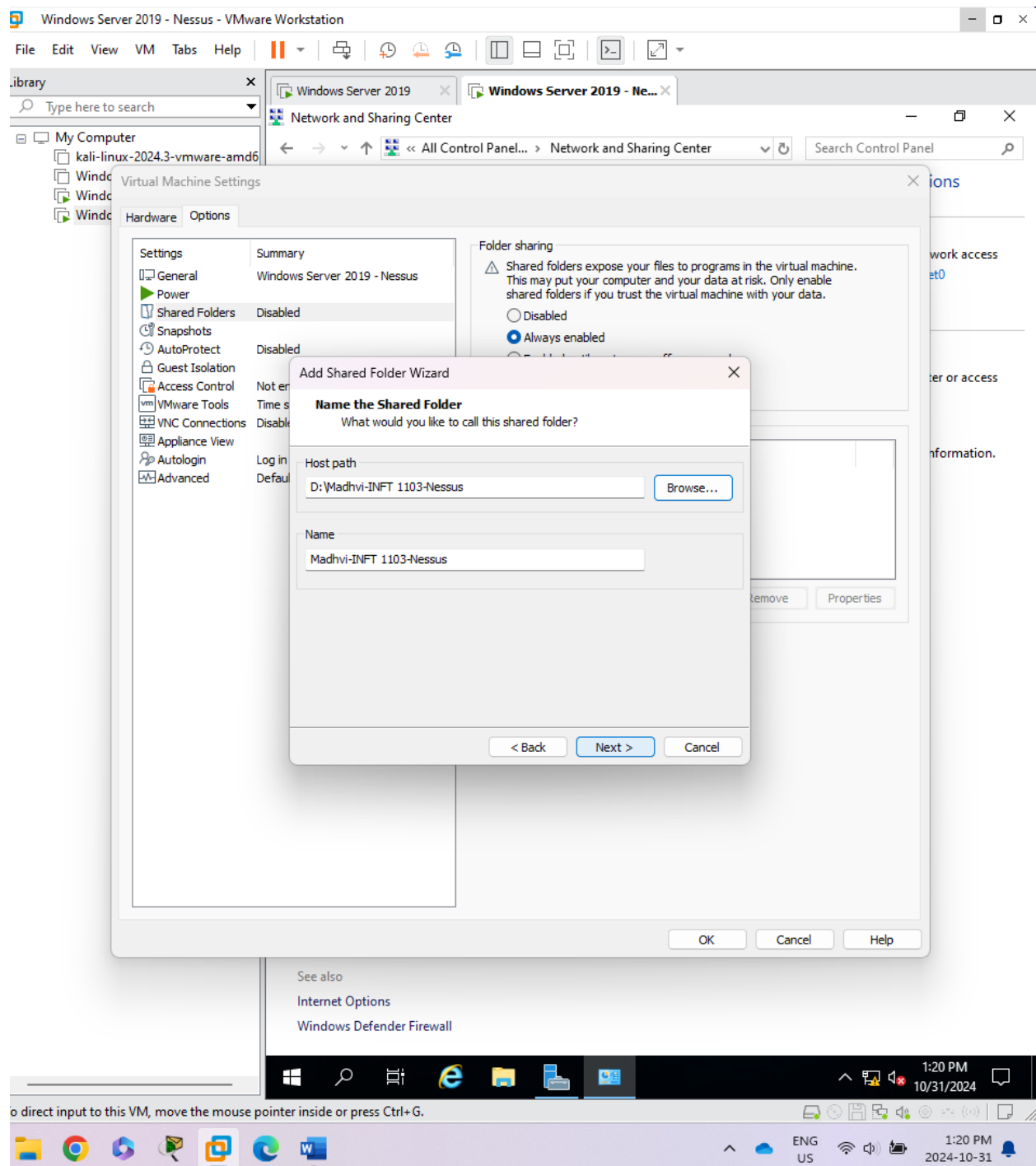
13. Create a new folder by clicking on the Make New Folder in a directory of your choice, in my own case, I created my folder in my Documents, name the folder “YourFirstname-INFT 1103-Nessus” and click OK

[Attach your screenshot]



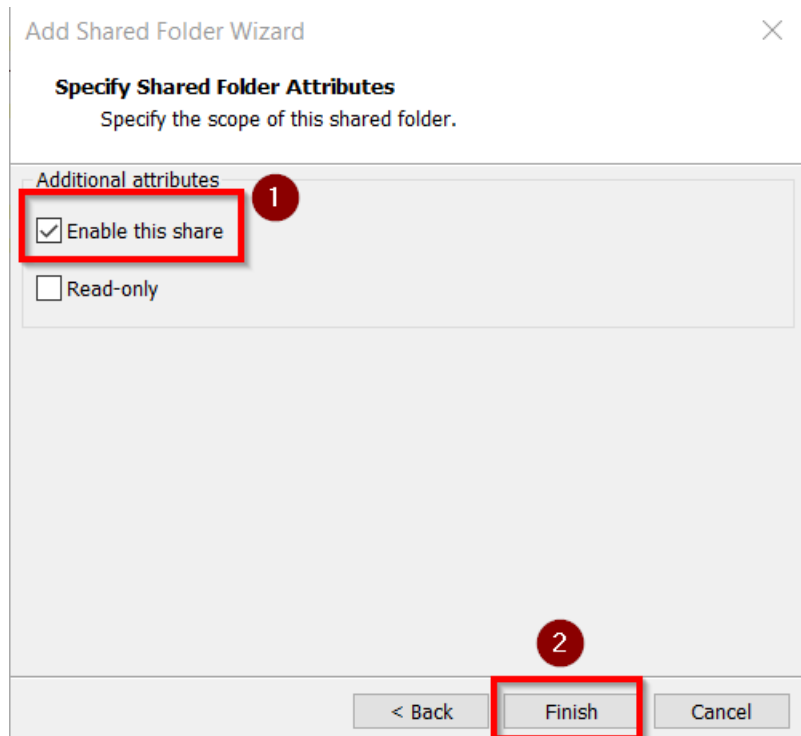
14. Our selected folder and name of folder would be populated. Then we click on Next >

[Attach your screenshot]



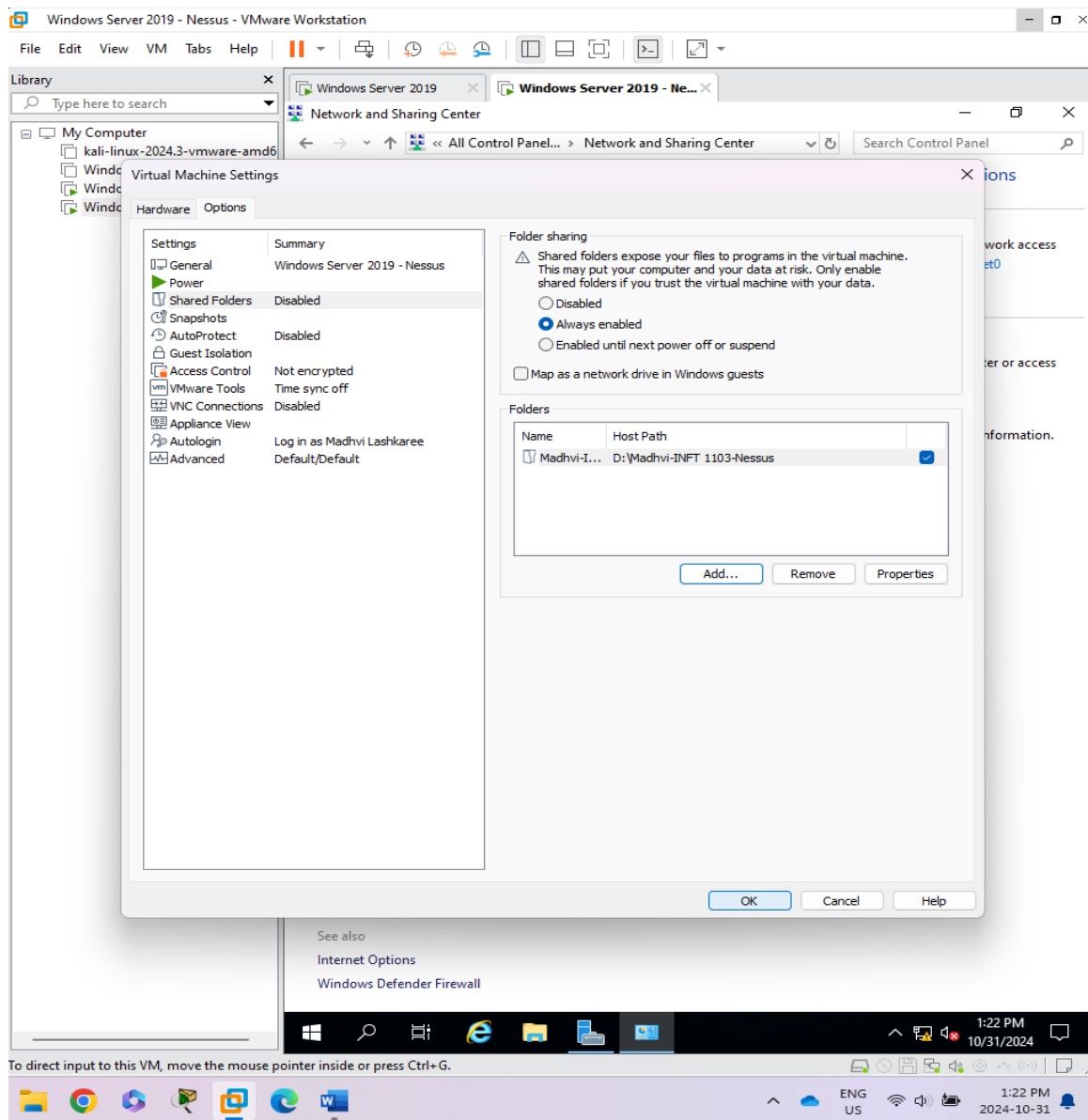
15. Make sure “Enable this share” is ticked

[Instructor's screenshot]



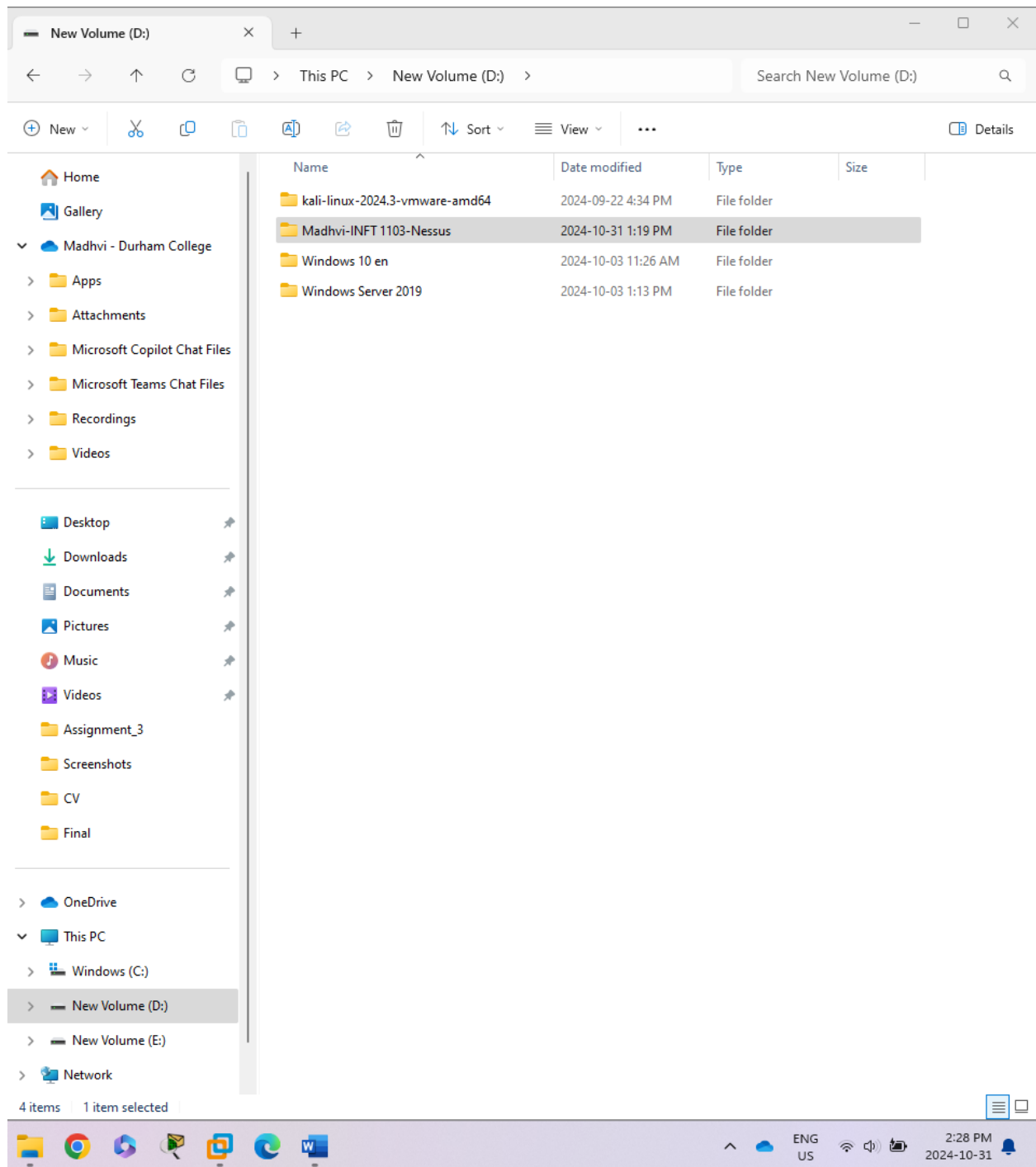
16. You would see your shared folder added and you click OK

[Attach your screenshot]



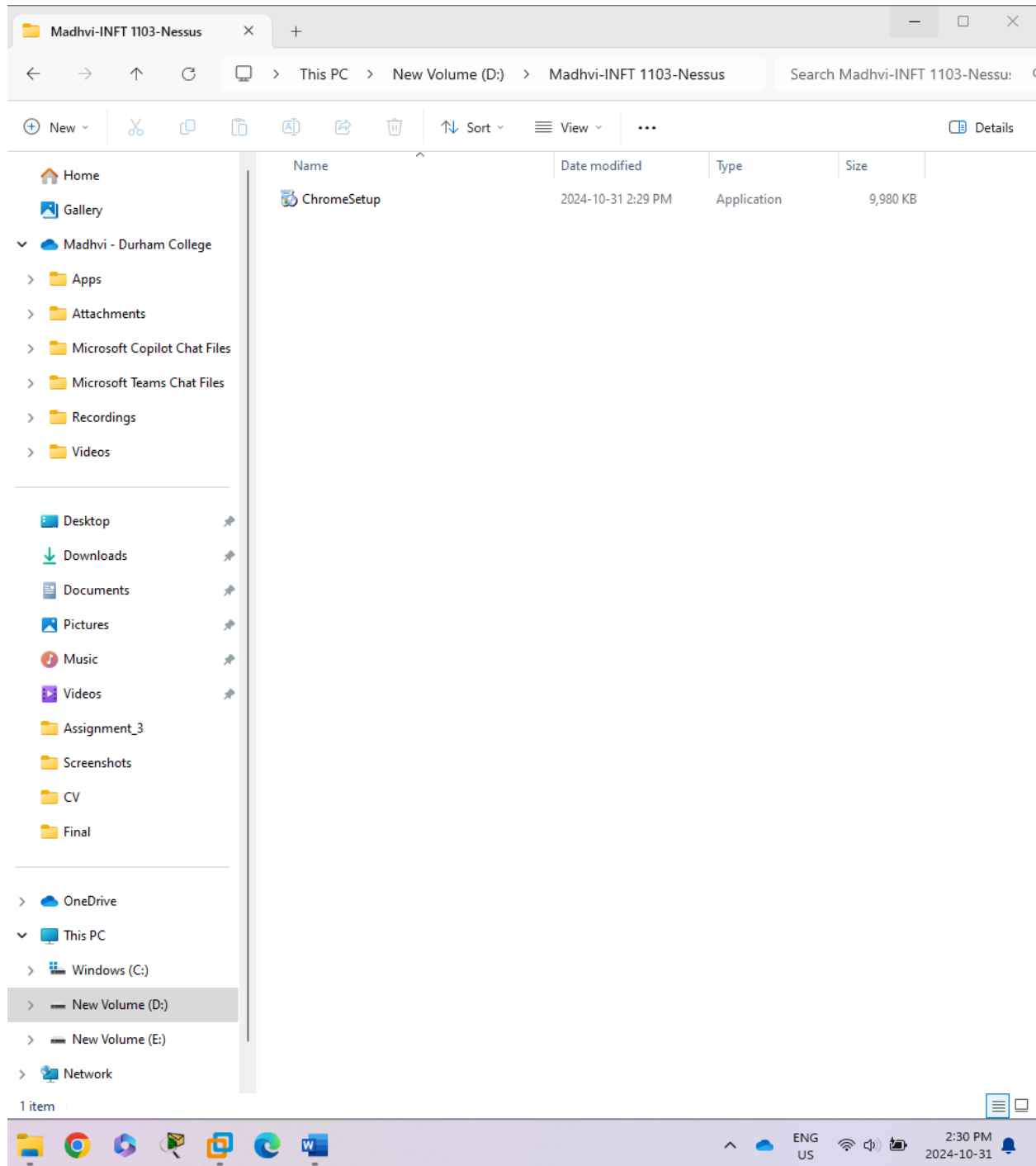
17. You should see you created folder in the folder you selected. In my own case, I saved it in my Documents folder

[Attach your screenshot]



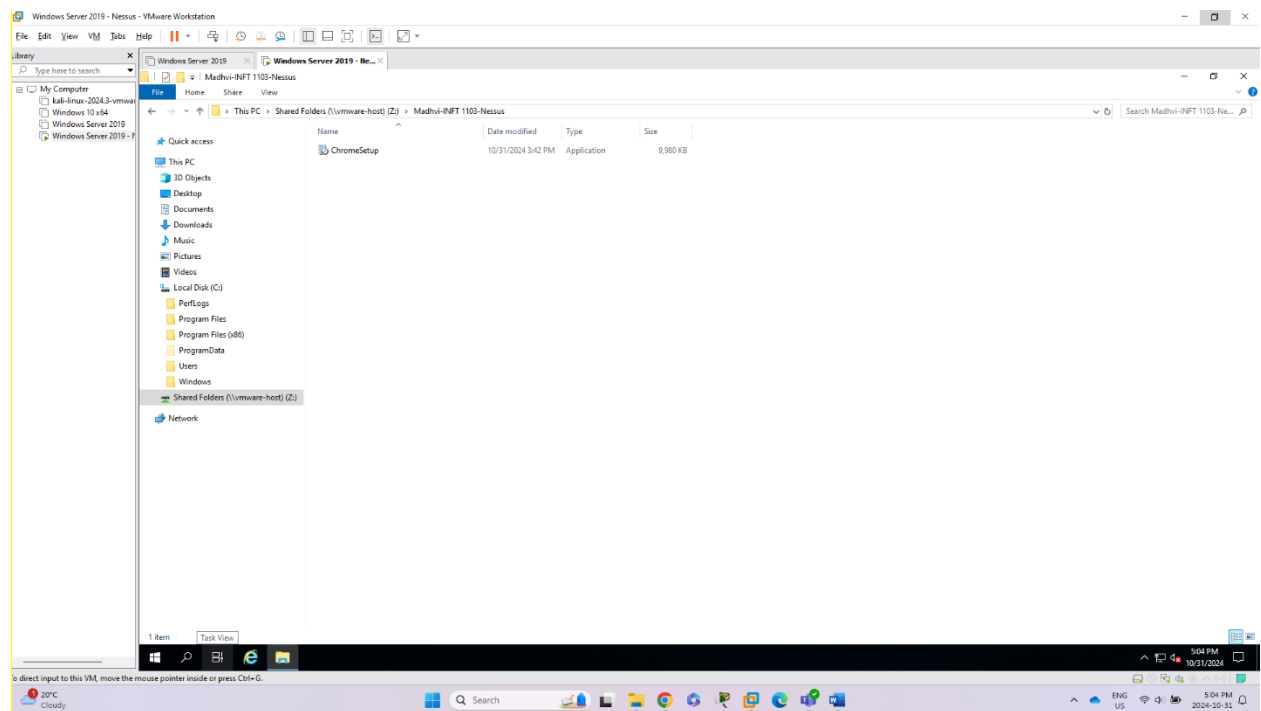
18. Download Chrome on your host laptop and copy it to your shared folder

[Attach your screenshot]



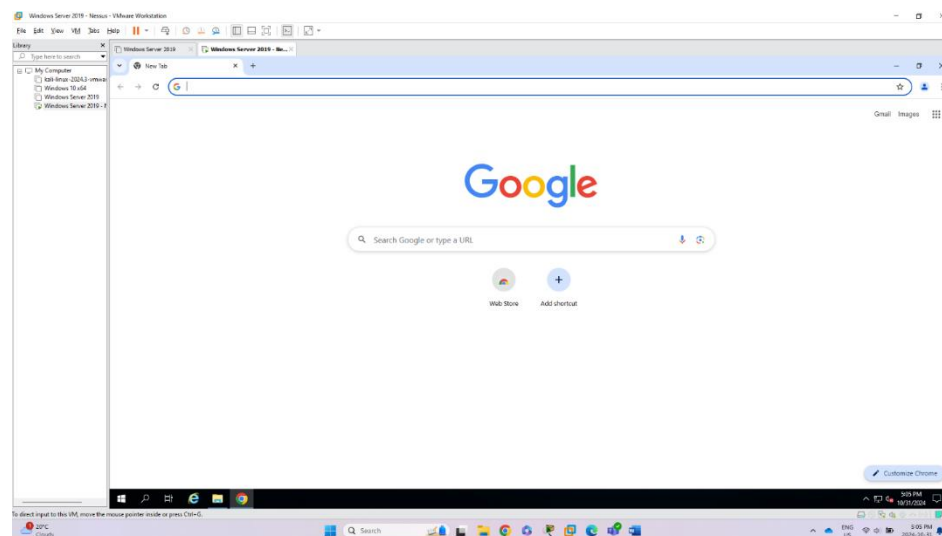
19. Turn on your windows server 2019 VM and check the “This PC” folder, you would see the shared folder from your host machine attached

[Attach your screenshot]



20. Open your shared folder to access your downloaded chrome and install it on your windows server 2019

[Attach your screenshot]

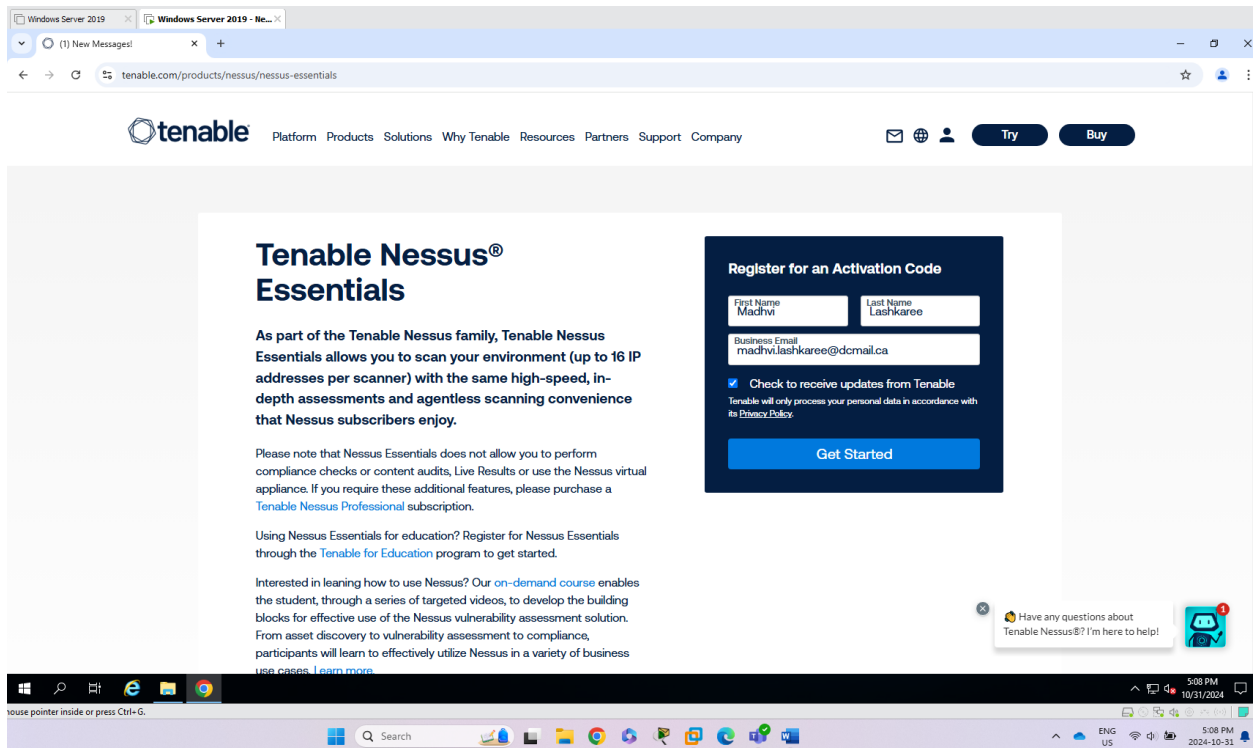


Download and Install Nessus

1. Go to the Nessus website on your windows server 2019 to register and get an activation Key, use your Durham college students email as the business email and you should get an activation code in your email in few minutes.

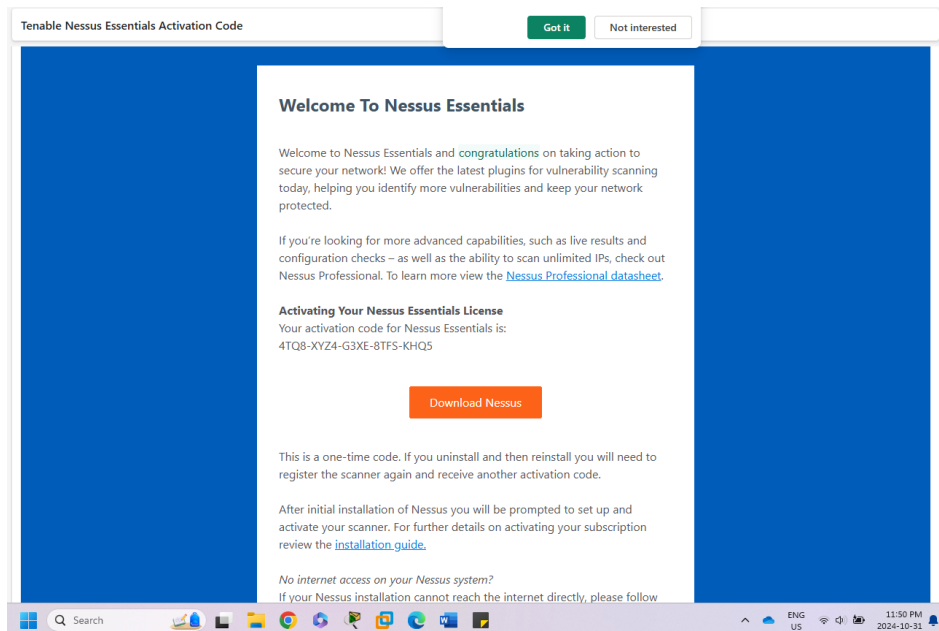
Nessus Website: - <https://www.tenable.com/products/nessus/nessus-essentials>

[Attach your screenshot]



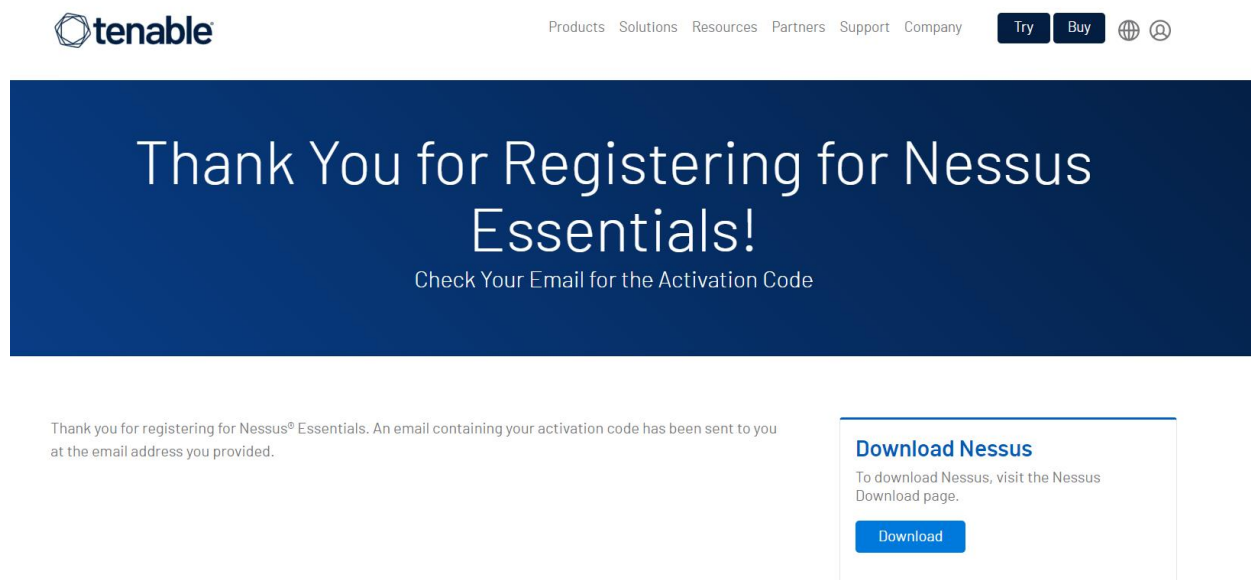
2. Registration key would be sent to your email after activation and email would look like this.

[Attach your screenshot]



3. After registering on the Tenable website, you would be taken to the download page. Or you can visit the Tenable website to download Nessus on your Windows server vm
Nessus Download website: - <https://www.tenable.com/downloads/nessus?loginAttempted=true>

[Instructor's screenshot]



4. In this lab, we will be downloading Nessus for windows server 2008, server 2012, server 2012 R2, 7, 8, 10, 11, Server 2016, Server 2019, Server 2022(64-bit). See screenshot below

[Instructor's screenshot]

 Nessus-10.1.0-x64.msi	Windows Server 2008 R2, Server 2012, Server 2012 R2, 7, 8, 10, 11, Server 2016, Server 2019, Server 2022 (64-bit)	84.2 MB	Jan 30, 2022	Checksum
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5. Accept the license agreement, Once Accepted, your file would be downloaded immediately

[Instructor's screenshot]

License Agreement

comply with applicable and reasonable security procedures provided Customer provides Tenable with such written procedures in advance. Some of the Professional Services may require Customer to have specialized knowledge or meet particular software or hardware requirements (for example, appropriate computers or appliances, stable Internet connection or up-to-date web browsers or operating system, etc.). If technical issues arise during the Professional Services, Tenable will use commercially reasonable efforts to resolve such issues, but will have no liability based on Customer's failure to meet technical requirements. Tenable will not provide any refund based on Customer's failure to meet these prerequisites.

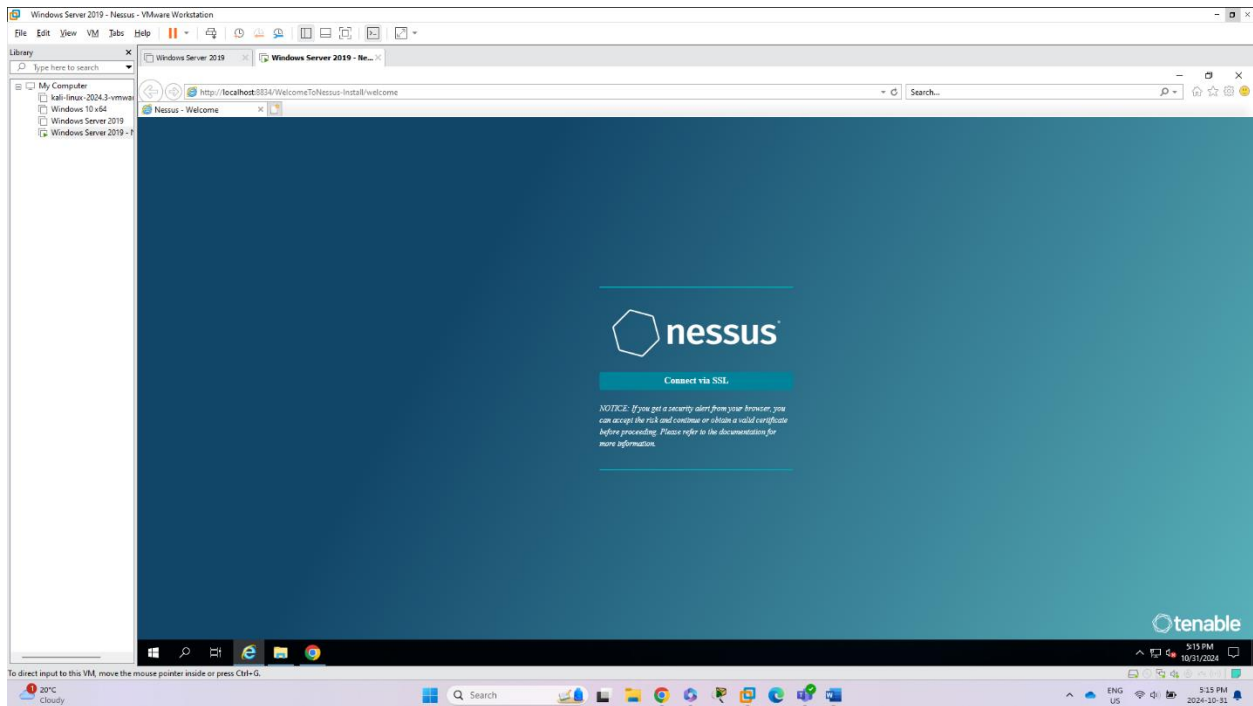
8. Changes. Either Party may request that a change be made to the Professional Services. Tenable reserves the right to charge a fee for any material changes to the Professional Services. No changes shall be binding unless executed by both Parties.

9. Non-Solicitation. During the term that Professional Services are being provided and for a period of one (1) year after their completion, Customers will not, either directly or indirectly, solicit for employment any person employed by Tenable or any of its Affiliates that have provided Customer Professional Services under this Agreement. For the avoidance of doubt, this restriction shall not prevent Customer from hiring based on a response to Customer's advertising in good faith to the general public a position or vacancy to which an employee or worker of Tenable responds, provided that no such advertisement shall be intended to specifically target Tenable personnel.

[I Agree](#) [Cancel](#)

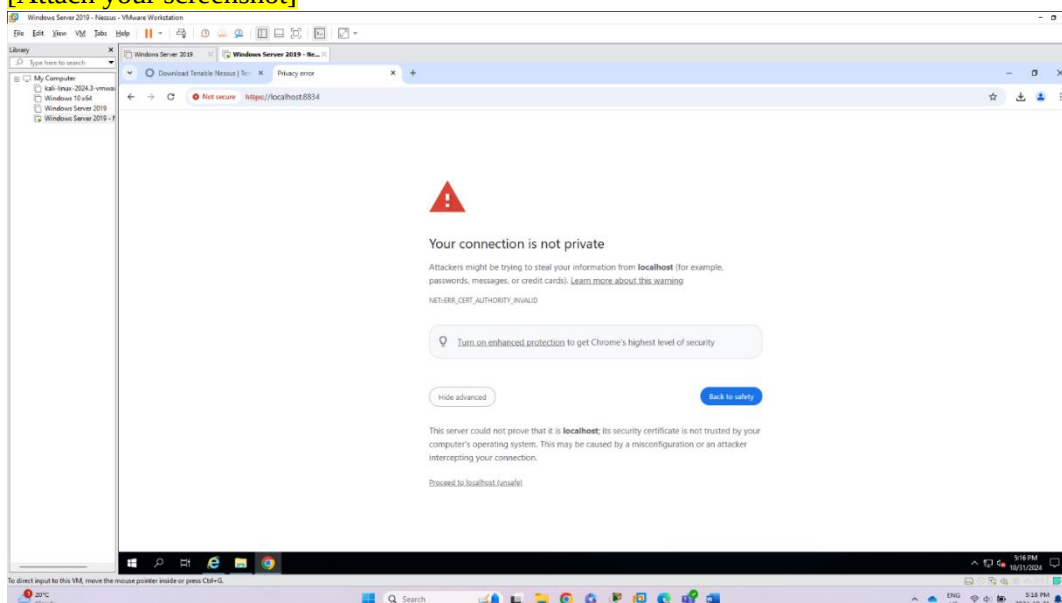
- Once downloaded, double click your Nessus file and start installing immediately. Follow the prompt for installation and click finish once installation is done

[Attach your screenshot]



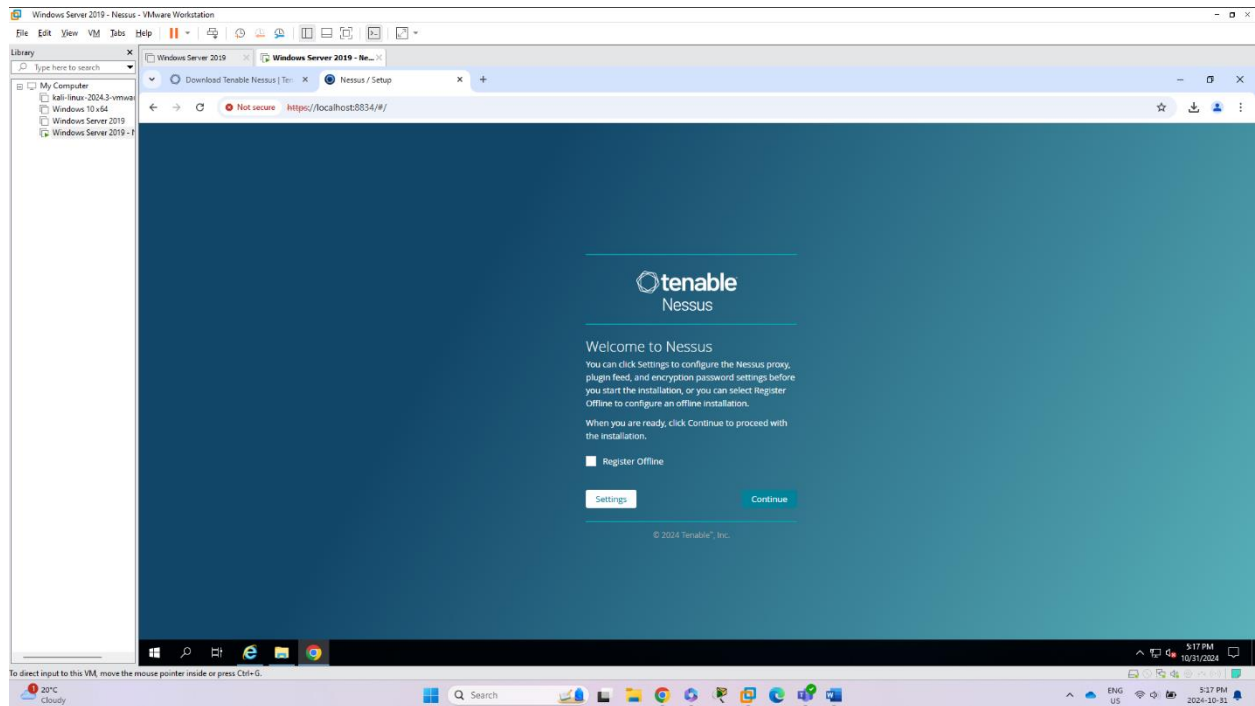
- Once installation is done, visit your chrome browser to access Nessus. You can use the url <https://localhost:8834> to visit Nessus. You would get a non-secure error in the browser click advanced and click proceed to local host unsafe

[Attach your screenshot]



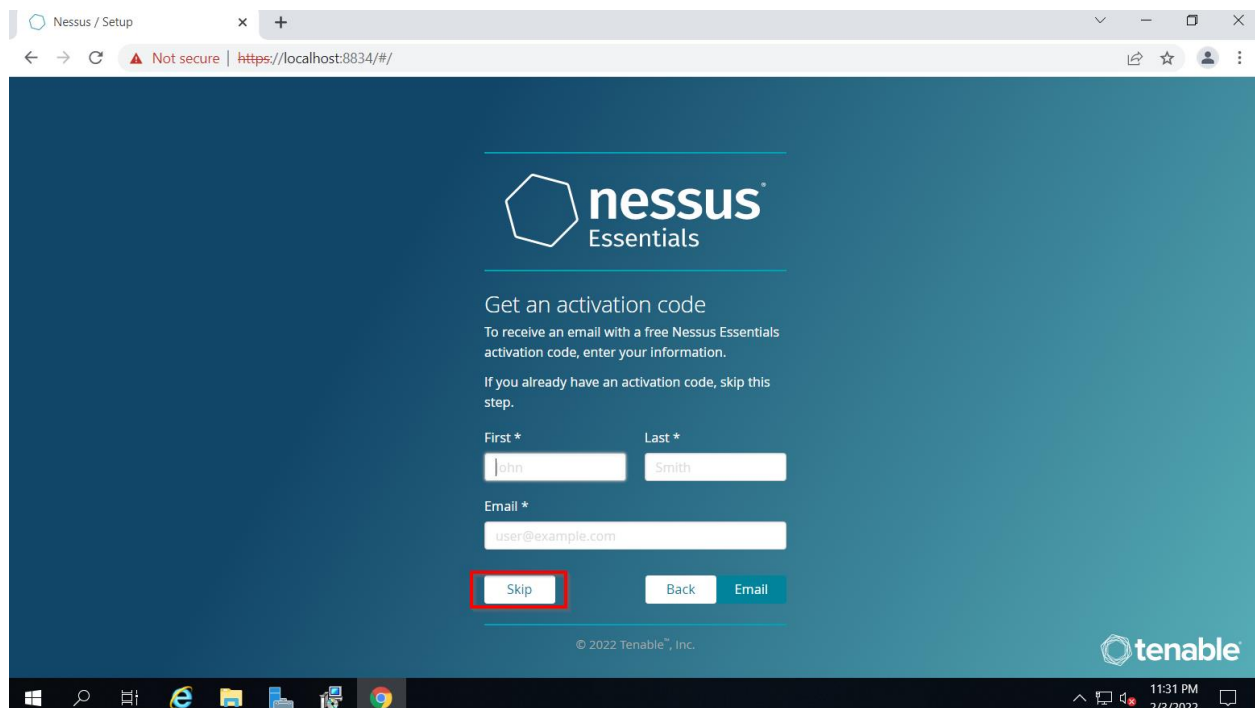
8. You should get the “Welcome to Nessus” Page. Select Nessus Essentials

[Attach your screenshot]



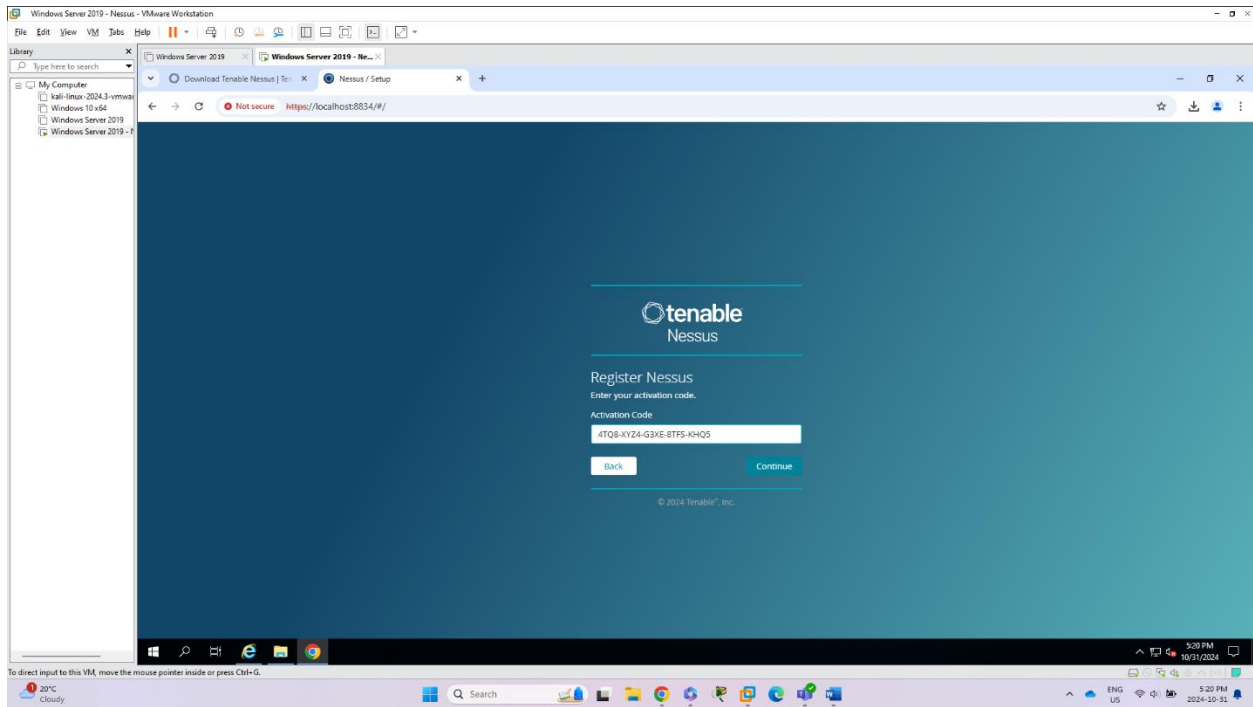
9. It would take you to the code activation page. Select skip since you have initially requested for one and already have your code sent to your email

[Instructor's screenshot]



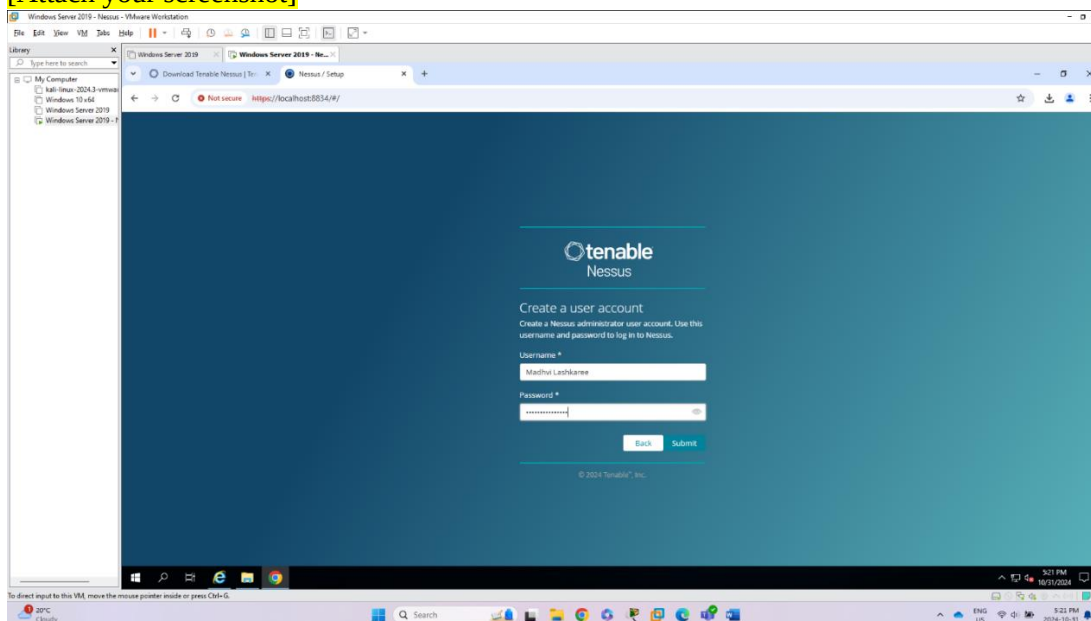
10. The Register Nessus screen would pop up, copy your Nessus code from your college email and paste it in the activation code box.

[Attach your screenshot]



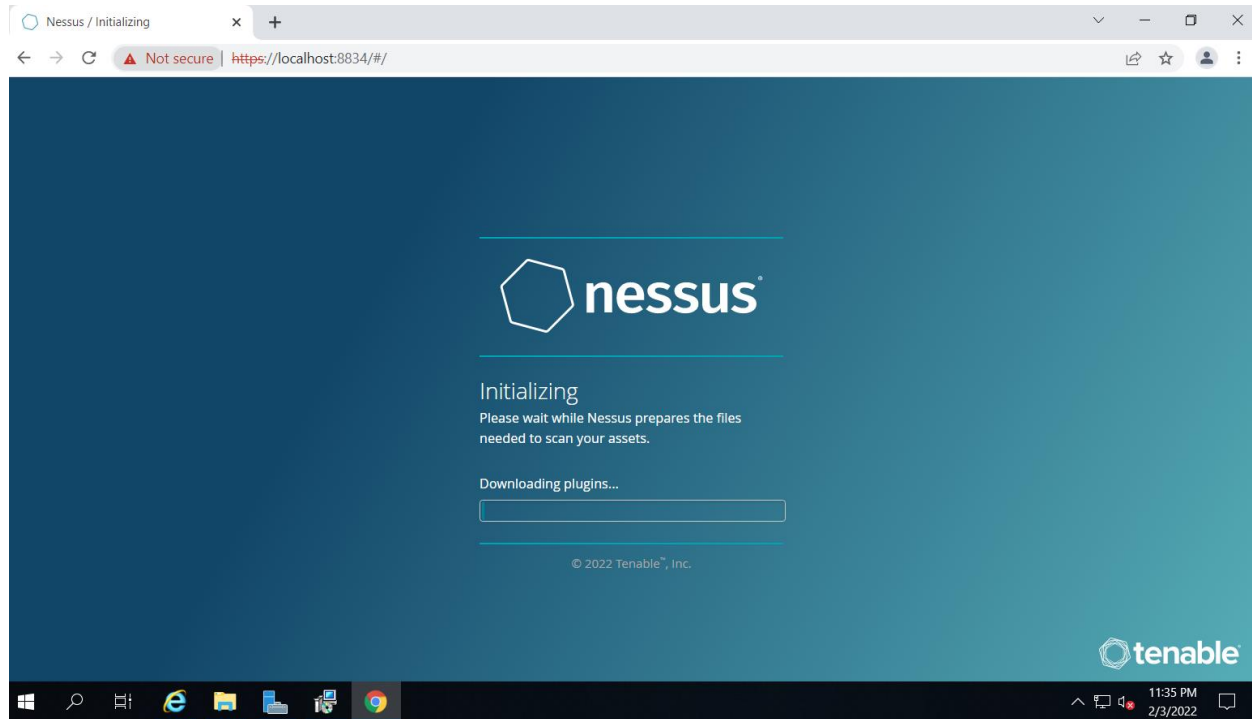
11. Next step is to create a new user we would use to login to Nessus

[Attach your screenshot]

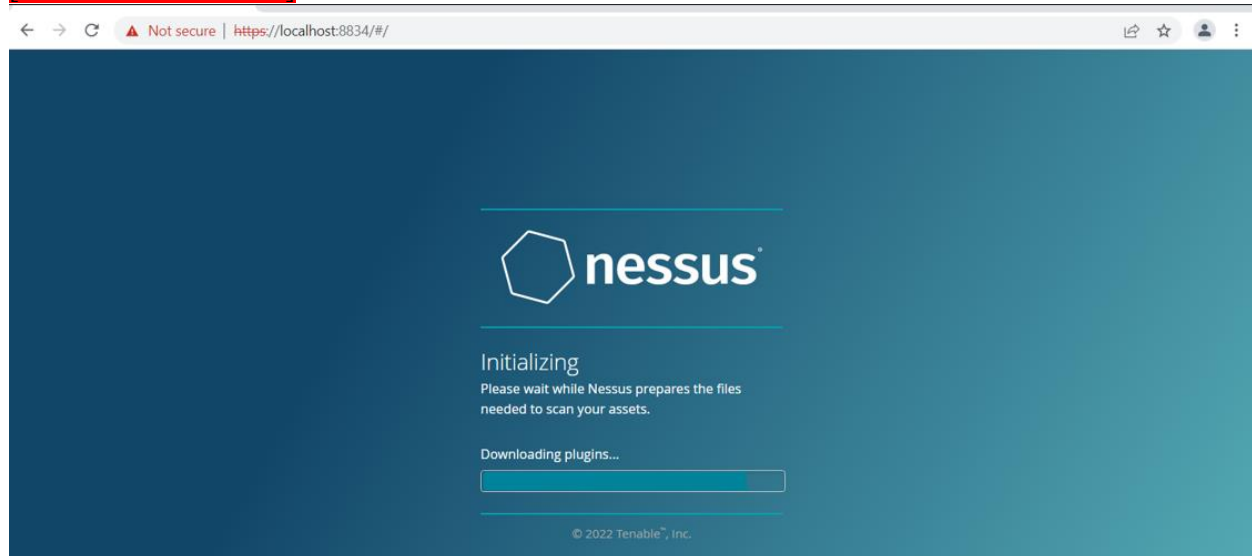


12. The next screen would be our Nessus installing. This would take some time, so sit back relax and have a coffee while your installation is running

[Instructor's screenshot]

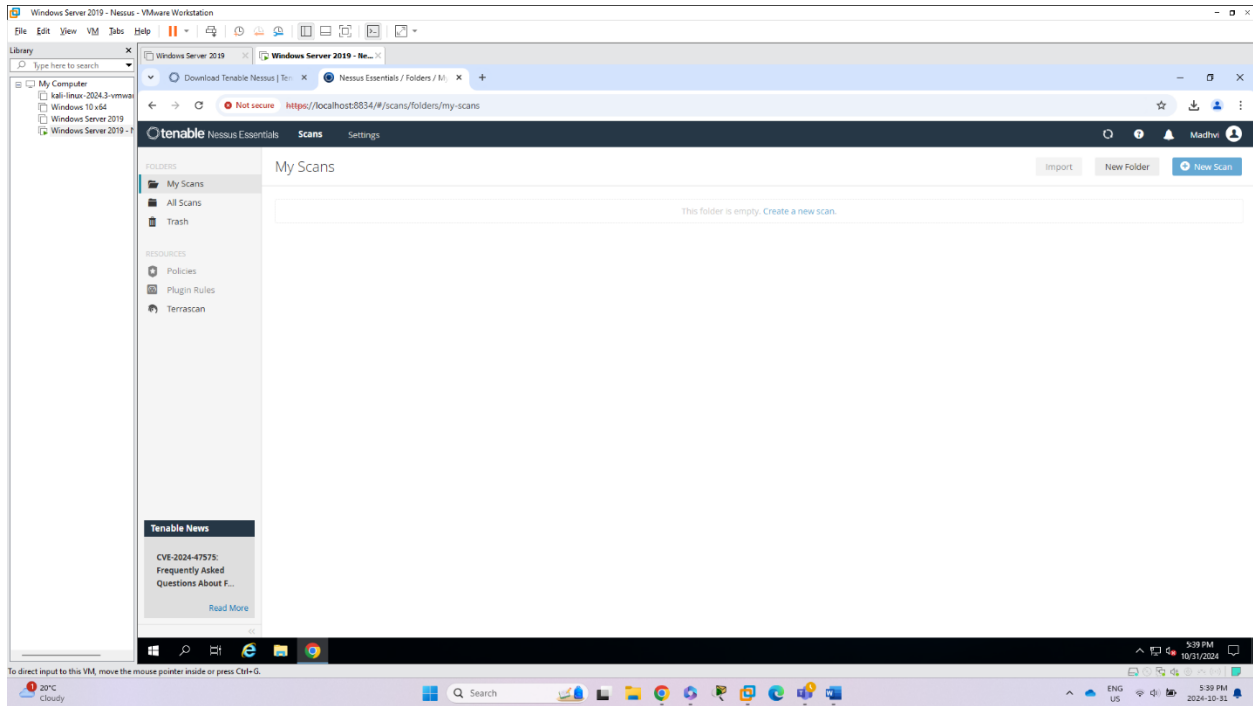


[Instructor's screenshot]



13. After Installation, you would see the Nessus welcome Page.

[Attach your screenshot]



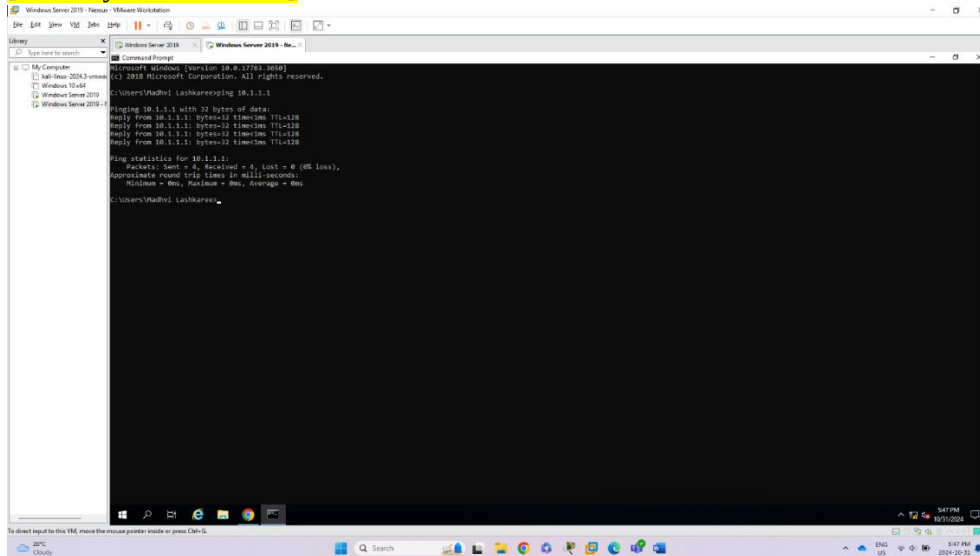
Scan your window 2016 Domain Controller using Nessus Host Discovery Scan

We would run a Host discover scan and a vulnerability scan

Before running a scan, ensure your windows 2019 server is able to ping your windows 2016 Domain controller

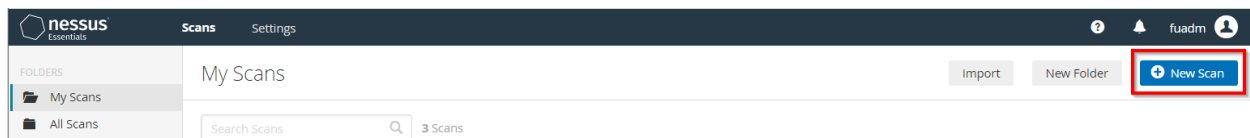
1. Open command line and ping your Domain controller

[Attach your screenshot]



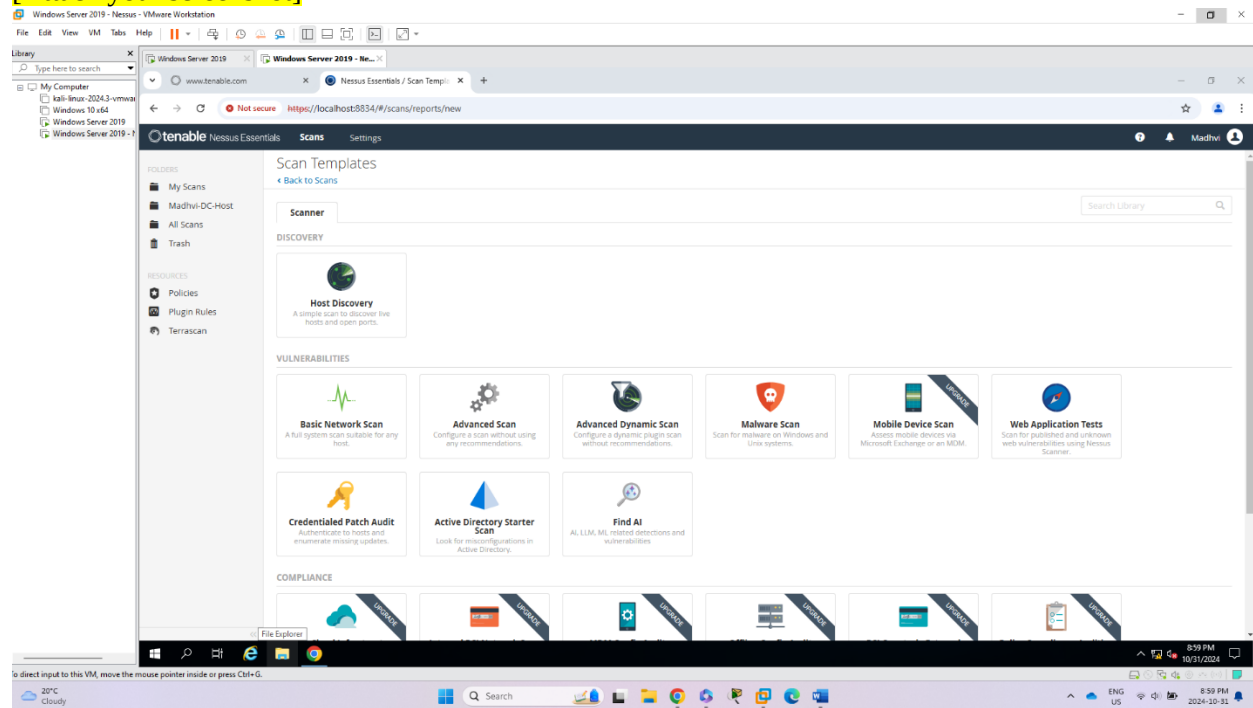
2. Go over to Nessus and click New Scan

[Instructor's screenshot]



3. Click Host discovery scan

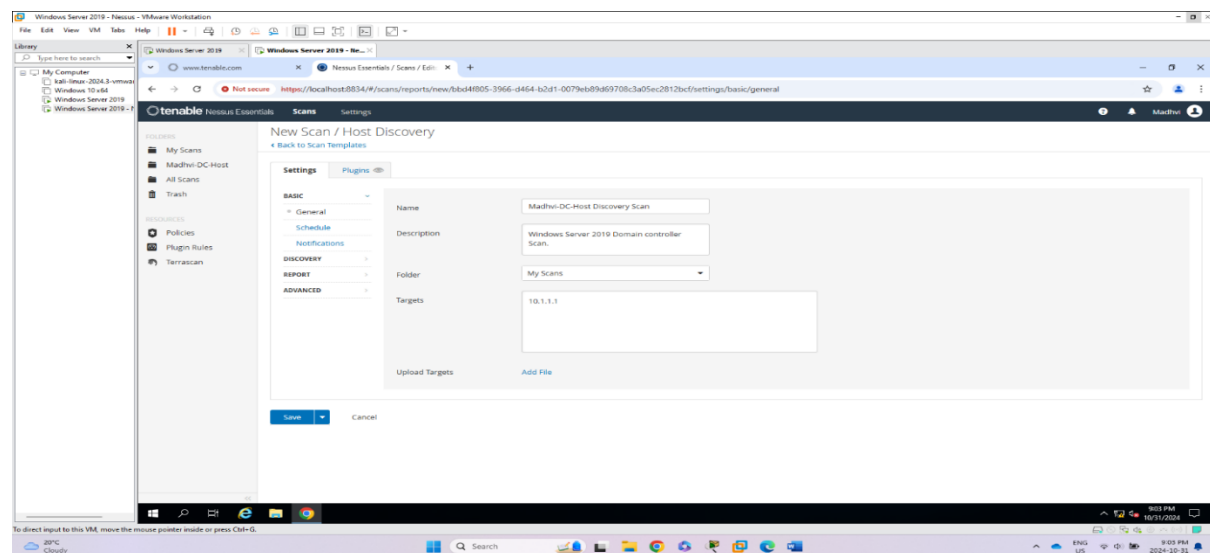
[Attach your screenshot]



4. Fill in the details for your scan.

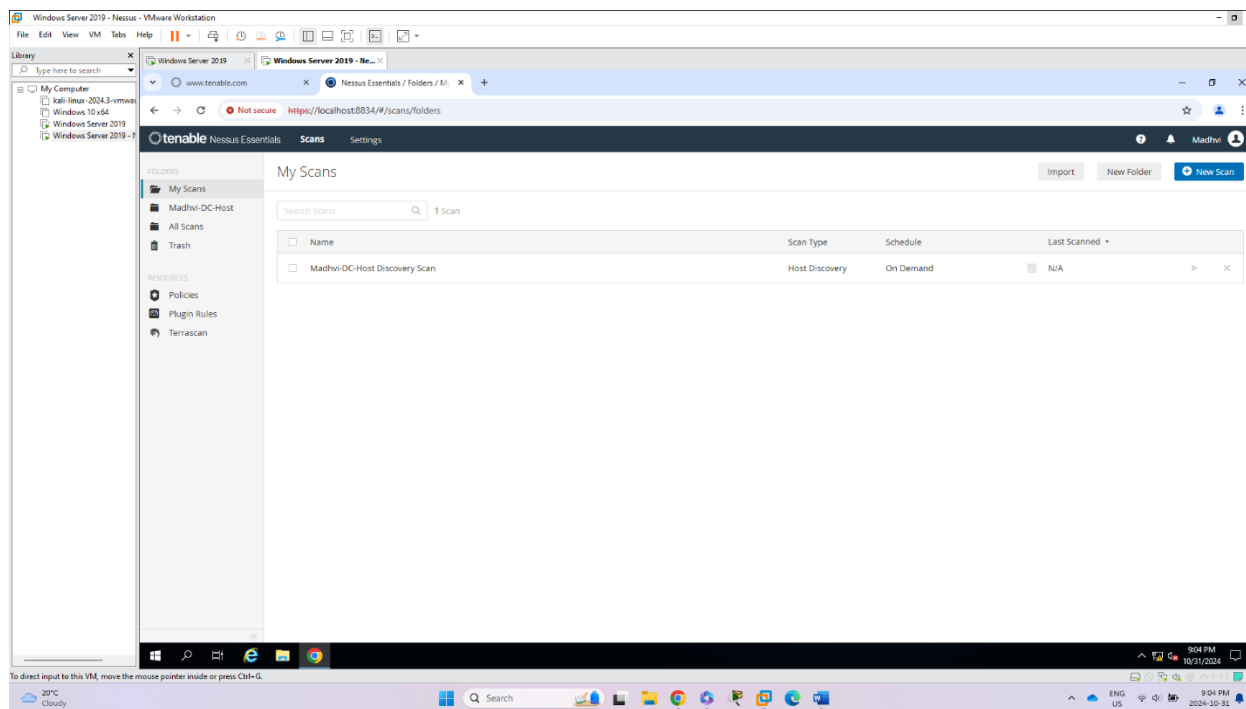
- Name your scan “YourFirstname-DC-Host Discovery Scan”
- Input your Domain controller IP “10.1.1.1”
- You can leave other functions as default
- Click save

[Attach your screenshot]



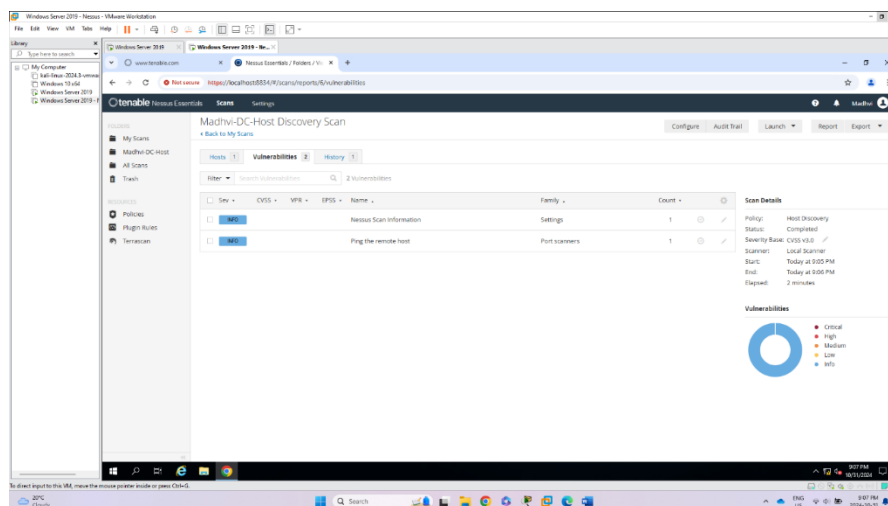
- Then run your host discovery scan against your domain controller by clicking on the play button

[Attach your screenshot]



- Your results should look like this.

[Attach your screenshot]



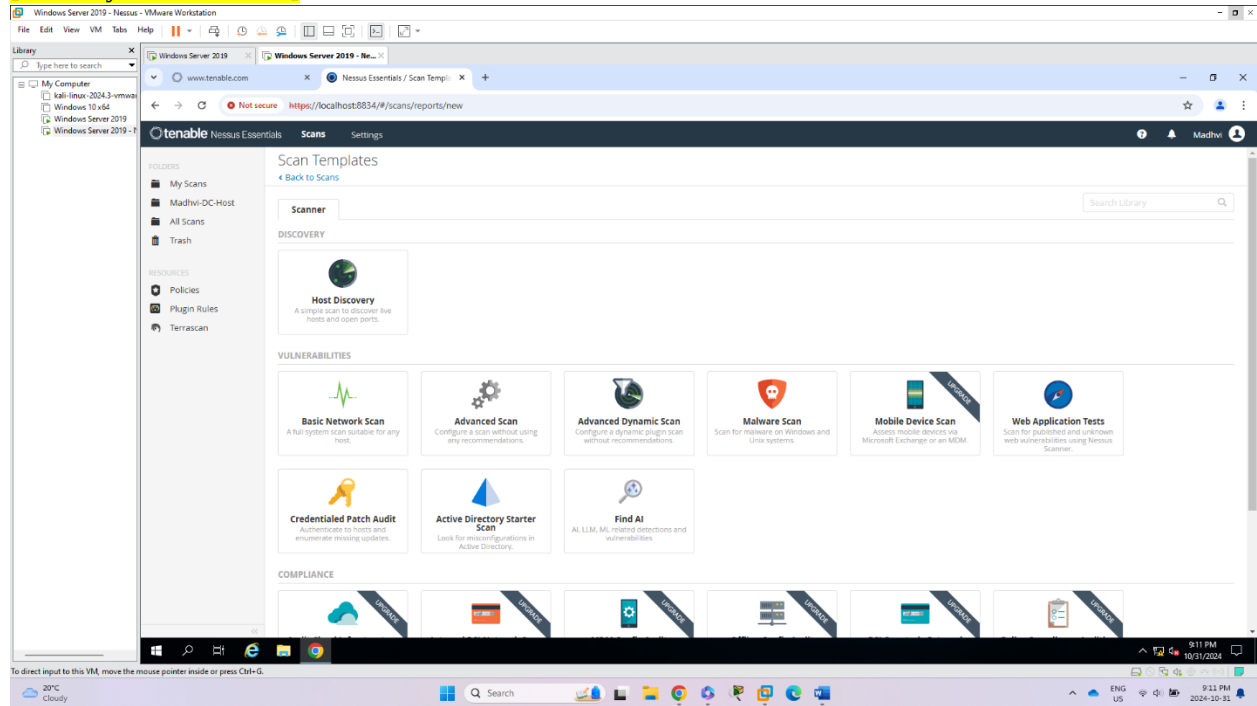
Good Job, you have gotten results from your host discovery scan

Vulnerability Scan

Once you get results from your host discovery scan, go ahead to run a vulnerability scan on our DC

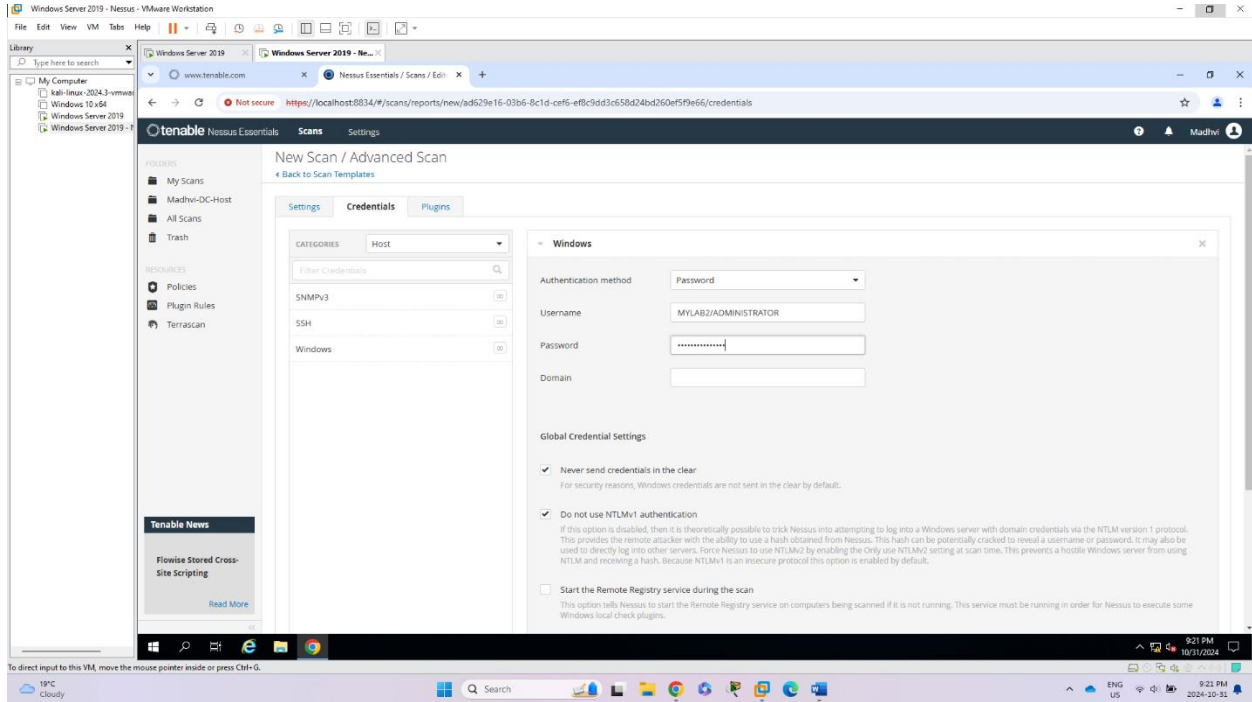
1. Click on Advanced scan

[Attach your screenshot]



- You should add your scan details. Your vulnerability scan name should be in the format “YourFirstName – DC – Vulnerability scan” .

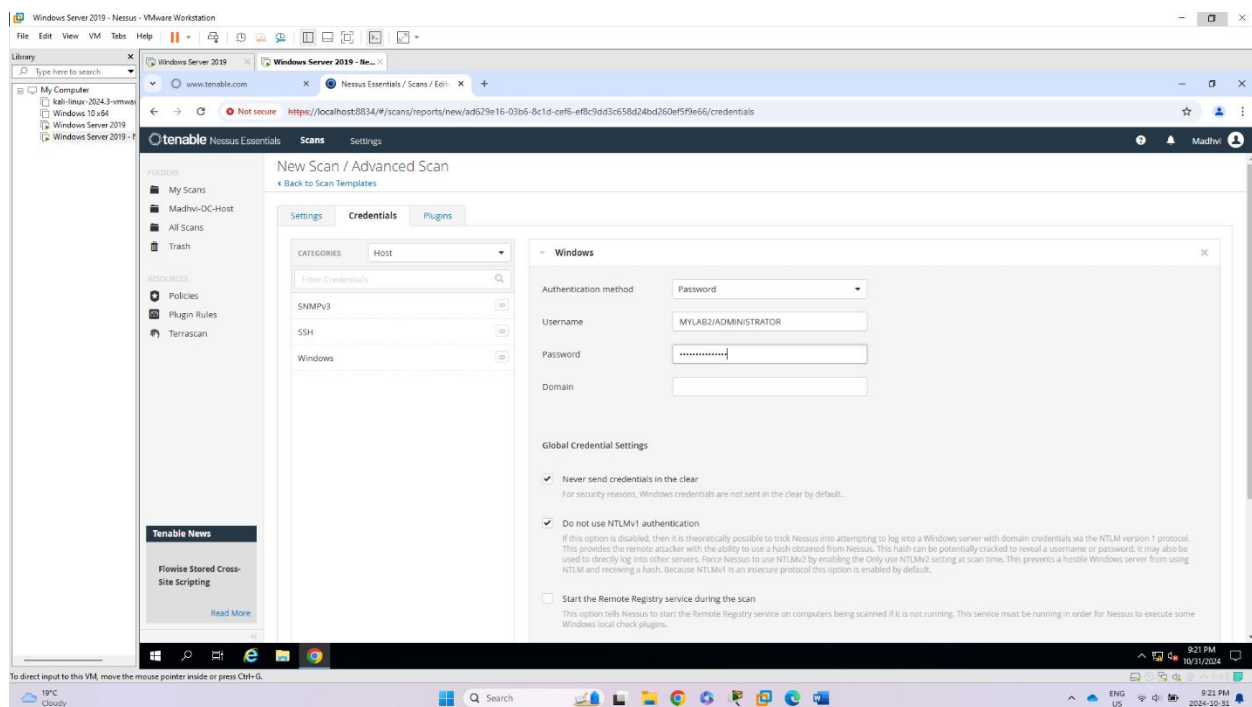
[Attach your screenshot]



- Next step is for us to add credentials in our scan. We would be running a credentialed scan. Click on credentials and click on windows. After doing this, you would input your windows server 2016 Domain controller credentials (username and Password), Nessus would need this credential to log into your windows server 2016 DC host for a more detailed scan. After inputting these details, click save

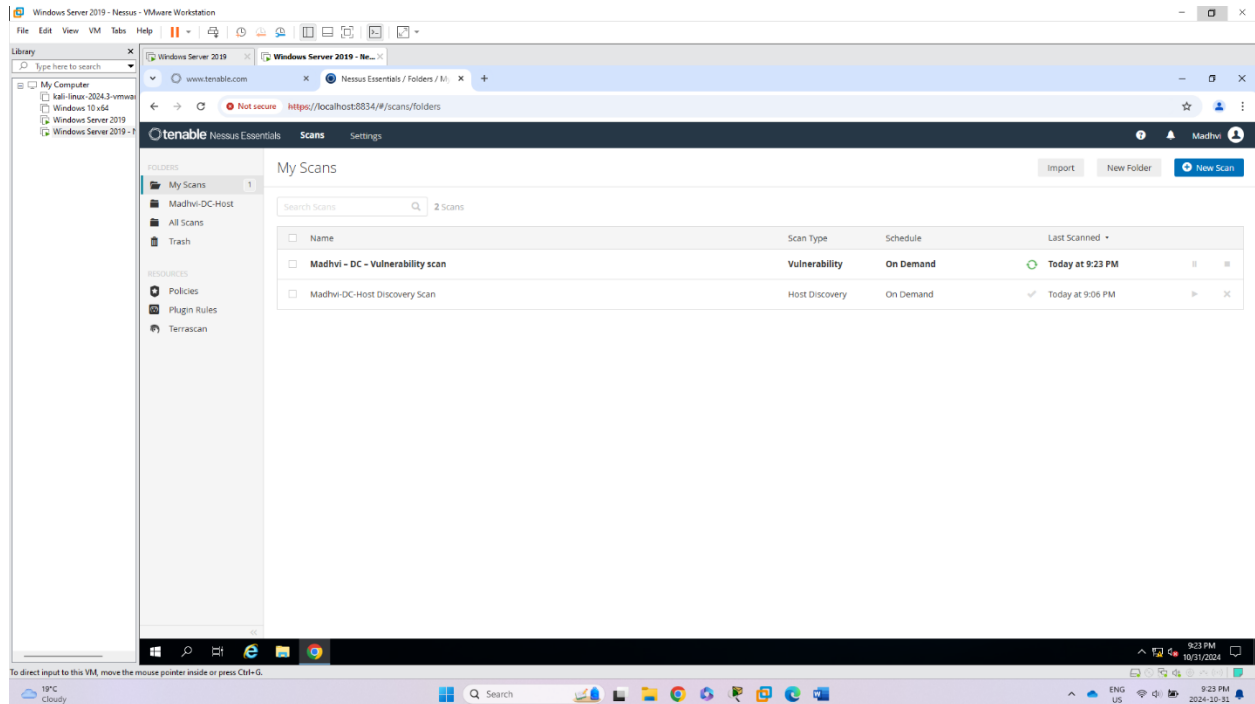
“NOTE make sure the credentials you are adding to your credentialed scans are of your windows server domain controllers. You must also make sure your credentials for your windows server 2016 domain controller are correct, if you do not get enough vulnerabilities, this is because your credentials are not correct”

[Attach your screenshot]

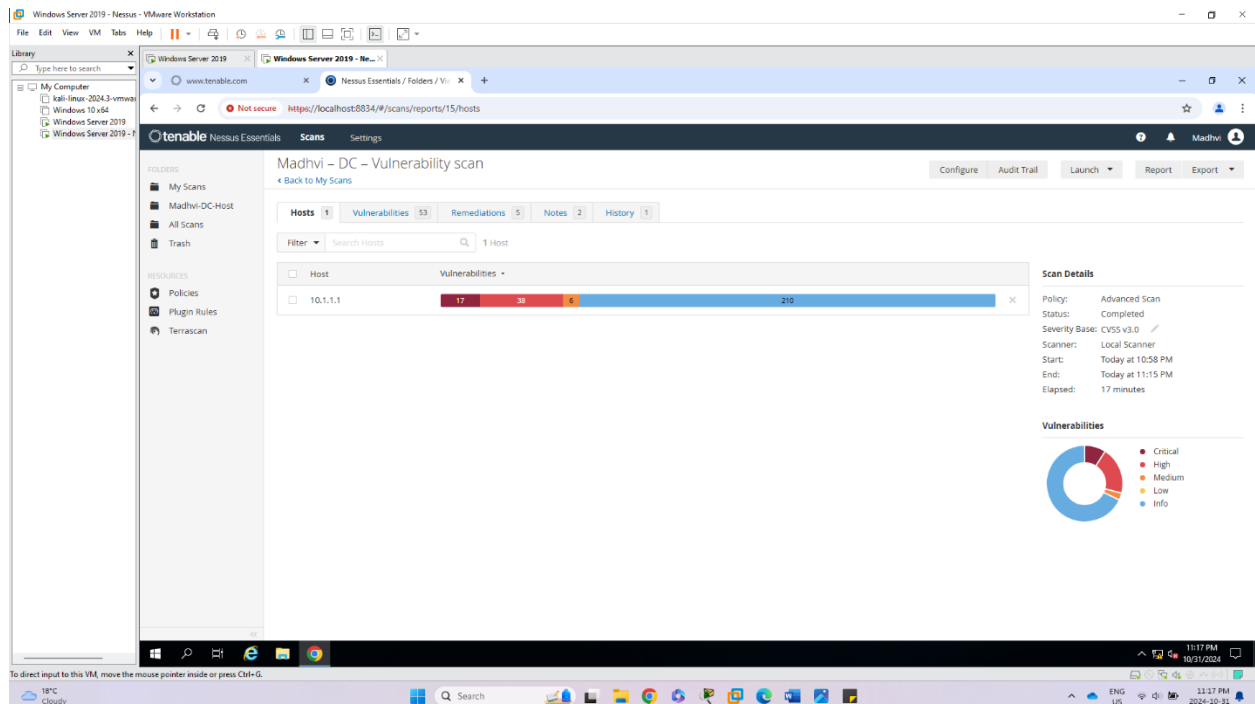


- Run a vulnerability scan on your windows server 2016 domain controller, this should take some time. You should get results after running the scan.

[Attach your screenshot] (screenshot showing scan running)



[Attach your screenshot] (Screenshot showing results)



[Attach your screenshot] (Showing vulnerabilities)

The screenshot shows the Tenable Nessus Essentials interface for a vulnerability scan titled "Madhvi - DC - Vulnerability scan". The interface includes a sidebar with navigation options like "My Scans", "Policies", and "Terrascan". The main content area displays a table of vulnerabilities with the following columns: Sev, CVSS, VPR, EPSS, Name, Family, and Count. The table lists three vulnerabilities, all of which are "MEDIUM" severity. A donut chart on the right side of the interface shows the distribution of vulnerability severities: Critical (red), High (orange), Medium (yellow), Low (green), and Info (blue). The chart indicates that the majority of vulnerabilities are of "Medium" severity.

Sev	CVSS	VPR	EPSS	Name	Family	Count
MEDIUM	...	...	...	Microsoft .NET Framework (Multiple Issues)	Windows : Microsoft Bulletins	10
CRITICAL	...	...	...	Microsoft Windows 10 1809 (Multiple Issues)	Windows : Microsoft Bulletins	5
MEDIUM	...	...	...	Microsoft System Center Endpoint Protection (Multiple iss...	Windows	4

Results per page: 50

Showing: 51 to 53 of 53

Scan Details

- Policy: Advanced Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: Today at 10:58 PM
- End: Today at 11:15 PM
- Elapsed: 17 minutes

Vulnerabilities

- Critical
- High
- Medium
- Low
- Info

Part C: Lab Report

Evaluation – Nessus

Create an Executive Summary from the Nessus findings

1. Executive Summary is written for executives not IT Professionals (necessarily)
2. Should have the following sections: 1. Engagement Summary, 2. Summary of Activities, 3. Summary of Findings, 4. Remediation strategies, 5. Next Steps.
3. Findings should have a description section, impact section, reference section and remediation section.
4. Description section should explain what it is, where it comes from. Impact should explain why it's bad and what the risk is.
5. Should be in your own words. Please **Do NOT copy** the tool's explanations
6. Remediation's should be actionable, executable and useful, not 'patch the out-of-date version' for example.
7. Include 3 detected vulnerabilities in your executive summary. (2 Critical, 1 High or medium)

CREATE YOUR
VULNERABILITY REPORT
IN A DIFFERENT
DOCUMENT USING A
GOOD TABLE OF CONTENT
AND TEMPLATE

Things to Explore:

You are welcome to explore beyond the mandatory requirements if you wish.

General submission requirements

- Include an opening comment with your full name, date, and a short description.
- **Marks distribution:**
 - a) 31 Screenshots each 1 mark = 30 marks
 - b) Video Demonstration = 5 marks
 - c) Lab Report = 5 marks
- **Do not alter the sequence of steps of this document. Do not delete the words [Attach the screenshot] anywhere from the assignment. Keep the numbering same. Paste the screenshots wherever it's being asked- do not use screenshots of professor's from lab manual. They are for your reference only and would be considered an academic integrity breach. You should be not even submitting another student's screenshot- Academic Integrity alert.**
- **Submit a video recording (sharing screen) briefly explaining the steps taken to perform the lab and show each of the VM running. Execute the commands in your VMs to show the domain and network configuration. If the video recording stating the above requirements are not submitted, you will be eligible to get marks out of 60% of Lab1 marks.**